BANK OF THAILAND

MONETARY POLICY REPORT

DECEMBER 2017

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Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
December 2017

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Porametee Vimolsiri	Member
Mr. Sethaput Suthiwart-Narueput	Member
Mr. Kanit Sangsubhan	Member
Mr. Subhak Siwaraksa	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for the medium term and for 2017. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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Executive Summary

Monetary Policy Conduct in the Fourth Quarter of 2017

Overall economic growth gained further traction in the fourth quarter of 2017 driven mainly by the external sectors, but the underlying strength of domestic demand and household purchasing power, headline inflation that remained below target, and pockets of risks to financial stability still warranted monitoring. Against this backdrop, **the Committee had to weigh between sustaining economic growth in order to attain the objective of price stability and preserving financial stability in formulating the most appropriate course of policy action. The Committee also assessed the benefits and costs of different policy options and unanimously voted to keep the policy rate unchanged at 1.50 percent at the meetings on November 8 and December 20, 2017.** In deliberating their decisions, the Committee judged that accommodative monetary policy stance would still be necessary to support the continuation of economic growth. The current policy interest rate remained appropriate in that it would facilitate sufficiently accommodative financial conditions to support economic growth and foster the return of headline inflation to target expected during the first half of 2018.

Looking ahead, the Committee deemed that monetary policy accommodation should be maintained for some time to support more robust economic growth and foster the gradual return of headline inflation toward the medium-term target. The Committee would closely monitor inflation developments and assess structural factors affecting inflation dynamics. The Committee would stand ready to utilize available policy tools to facilitate the return of headline inflation to target in an appropriate time, while seeking to achieve sustainable growth and maintaining financial stability. The Committee also deemed it necessary to develop a process to monitor and assess financial stability risks that could be systematically incorporated in the conduct of monetary policy.

Monetary Policy Target in 2018

The Committee and the Minister of Finance concurred that headline inflation of 2.5 ± 1.5 percent should be the target for the medium term and for 2018. The Cabinet approved the proposed target on December 19, 2017, as this level of inflation would facilitate economic growth to be in line with potential as well as maintain the country's competitiveness. In addition, the tolerance band should be appropriate to provide cushion against shocks that might cause inflation to deviate from target in the short term.

Assessments of the Economic, Inflation, and Financial Stability Conditions and Outlooks as the Basis for Policy Formulation

1. Global Economy

The global economy was projected to continue expanding and remain a driver of Thai exports in the period ahead. Advanced economies expanded further mainly due to consumption and stronger labor markets. Moreover, the U.S. tax policy reform, signed into law toward the end of December 2017, would help stimulate the U.S. economy going forward. The euro area recorded stronger expansion on the back of consumption and exports. In addition, accommodative financial conditions, improved consumer confidence, and a recovery in the labor market would help support consumption spending going forward. Japan continued to expand on account of exports and manufacturing, with consumption expected to increase following improved consumer confidence and labor market conditions. Meanwhile, the Chinese economy was expected to gain further traction despite some slowdown because of financial stability measures implemented by the authorities. However, improving global demand and the government's additional financial measures to support specific groups would facilitate economic growth going forward. Other Asian economies were expected to grow following improvements in exports which would underpin employment and household consumption in the period ahead. **The Committee therefore revised up the growth forecast for Thailand's trading partners to 3.8 percent in 2017.** However, there remained certain risks that warranted monitoring in 2018. These included uncertainties surrounding U.S. economic policies such as trade and infrastructure policies as well as geopolitical risks that could undermine the economy and financial markets.

Most central banks maintained accommodative monetary policy stance. Nonetheless, some central banks raised their policy rates such as the Federal Reserve (Fed), the Bank of England, and the Bank of Korea. The Fed commenced its balance sheet reduction as previously announced and the European Central Bank announced a reduction of monthly bond purchases. Moreover, a number of central banks in Asia started to signal changes in

their monetary policy direction, as improved economic growth and a gradual rise in inflation facilitated monetary policy normalization the period ahead.

2. Financial Conditions and Financial Stability

Financial conditions remained accommodative. Short-term bond yields remained below the policy rate due to a decreased supply of short-term bonds. Meanwhile, long-term bond yields edged up on the back of an increase in the long-term bond supply, together with external factors following the progress of the U.S. tax policy reform which could lead to budget deficits. Consequently, U.S. government bond yields were expected to rise and could lead to higher Thai government bond yields. Meanwhile, interest rates on new loans (NLR) remained stable at a low level after trending downward earlier. Overall private credit expanded driven mainly by household loans especially mortgage and auto leasing loans. Corporate loan growth, on the other hand, slowed down especially in the manufacturing and trade sectors in part due to debt repayments of large corporates. However, SME loan growth started to pick up in several sectors and working capital loans to export-oriented businesses continued to expand. Regarding the exchange rates, the baht remained largely unchanged. The real effective exchange rate (REER) appreciated somewhat consistent with improvements in economic fundamentals and was broadly in line with REERs of other countries.

Financial stability remained sound but there remained pockets of risks that warranted monitoring. These included, first, debt serviceability of SMEs and low-income households as the positive spillovers from the economic expansion did not yet broaden out, as reflected in deterioration in credit quality. Second, the search-for-yield behavior that could lead to underpricing of risks needed to be monitored, for example, a continued expansion in foreign investment funds (FIF) that were concentrated in some countries as well as increased investments in riskier assets of some saving cooperatives. Third, there was an oversupply of property developments in some areas, such as condominium units along the MRT Purple Line which have a longer time to go, as well as developments regarding the launches of mixed-use real estate projects that would raise supply in the next 4-5 years.

3. Economic and inflation outlook

The Thai economy was projected to continue expanding and achieve 3.9 percent growth in 2017 and 2018, higher than previously forecasted in the previous quarter. The upward revision was on account of a continued improvement in merchandise exports and tourism, underpinned by growth of trading partners' economies. Private spending gradually expanded and began to be more broad-based. Fiscal impetus also supported growth.

Merchandise exports continued to expand across various product categories and almost all export destinations. The value of merchandise exports in 2017 was revised up to grow by 9.3 percent in tandem with economic expansion of trading partners. Such improvement was particularly observed among exports of electronics, auto parts, and processed agricultural products, thanks to global demand for electrical products and the relocation of production bases to Thailand of some products such as smart phones. Moreover, export prices, especially for commodities, increased in line with oil prices. Meanwhile, import value also rose following an increased demand for raw materials and intermediate goods as well as higher oil prices. However, the exports expansion in 2018 was likely to slow down somewhat due to prior acceleration, together with specific factors such as the rise in exports following the relocation of production bases of some products to Thailand observed this year. However, the monthly average value of merchandise exports was still expected to reach a record high in many years.

Exports of services in 2017 was slightly lower than expected for non-tourism receipts, while tourism remained strong. The projection of the number of foreign tourists was revised up to 35.6 million in 2017 and 37.3 million in 2018 due to several reasons: (1) the increasing number of Chinese tourists, both group and independent tourists, due to new direct flight routes, (2) the rising number of ASEAN tourists which was in line with regional economic growth, and (3) a rise in tourism spending per head thanks to global economic expansion and higher-spending tourists.

Private consumption would gradually expand in the period ahead, supported by several factors: (1) improvements in farm income from the previous year due to higher output, (2) improved earnings of workers, especially those in the high-income group, in export-related manufacturing and tourism sectors, (3) maturing debts from the first-car scheme, and (4) government measures such as the social welfare card project. However, the labor market had yet to fully benefit from the economic recovery, partly due to structural

changes with increasing adoption of automation especially in some industries with strong exports. Other factors included migration of labor from the manufacturing sector to the service sector, where productivity and wages were lower, and elevated household debt, particularly of low-income households, for which deleveraging could take some time. Private spending would thus remain modest and might not be sufficiently broad-based going forward.

Public spending remained an important economic driver. Both public consumption and investment expenditure continued to expand despite some unexpected delays in some investment projects, constrained by limited disbursement efficiency and heavy rain which affected construction. Investment projects of state-owned enterprises (SOEs) were mostly on track, although some projects were delayed. Nevertheless, an increase in overall investment budget for 2018 and the holdover of some SOEs investment plans from 2017 would contribute to higher public investment in 2018. However, the promulgation of the Public Procurement and Supplies Management Act B.E. 2560 might result in a delayed disbursement during the initial phase of some state agencies that were not accustomed to the new system such as local administrative organizations.

Private investment continued to recover but was projected to expand at a modest pace. Investment recovery was observed in various industries, which was consistent with expansion in private consumption and exports. However, there remained excess production capacity in some businesses. Nevertheless, infrastructure investment projects and the enactment of the Eastern Economic Corridor Act would be supporting factors for private investment going forward.

Headline inflation remained low due to supply-side factors but was expected to slowly rise. In recent periods, headline inflation was at a low level close to the previous estimate. This was as a result of supply-side factors, namely an increase in output due to favorable weather condition. Going forward, inflation would edge up slowly on the back of a gradual rise in demand-pull pressures, given the improved growth outlook together with the impact from an increase in excise tax. However, inflation might be held down by several factors, such as fresh food prices that would likely remain low thanks to technological advancements and increasing price competition. **The Committee therefore projected headline inflation to be at 0.7 and 1.1 percent in 2017 and 2018 respectively, while core inflation was projected to be at 0.6 and 0.8 percent in 2017 and 2018 respectively. Moreover, the Committee assessed headline inflation to return to the lower bound of the target within the first half of 2018.**

Risks to the growth projection were expected to be in balance with the likelihood that the Thai economy would achieve stronger growth than the baseline projection, given a better growth outlook of Thailand's trading partners due to U.S. economic stimulus measures, China's growth moderation that was orderly, and a continued recovery of Asian exports. Other upside risks included a larger-than-expected public spending following the speedup of infrastructure investment and accelerated spending of funds accumulated by local administrative organizations. On the downside, there were possibilities that the growth outturn might be lower than the baseline projection due to uncertainties pertaining to U.S. foreign trade policy, geopolitical risks, and risks of lower-than-expected domestic spending as improvement in purchasing power was not yet sufficiently broad-based.

Risks to the inflation projection were also assessed to be in balance. Upside risks that inflation might be higher than the baseline projection could come from higher crude oil prices on the back of global economic recovery, heightened geo-political risks, and minimum wages increases in 2018. However, on the downside, inflation might fall below the baseline projection from weaker-than-expected demand-pull pressures.

1. The Global Economy

Advanced economies continued to gain further traction mainly on the back of stronger consumption, labor market conditions and manufacturing. The gradual pick-up in investment would provide additional growth momentum in the period ahead.

The U.S. economy continued to expand on the back of improvement in private consumption, underpinned by robust consumer confidence, a stronger labor market, and sound household financial positions. Meanwhile, private investment also expanded following continued improvements in business confidence and gradual pick-up in corporate profits, which would be an important growth driver in the period ahead. In the third quarter, growth of the U.S. economy was more broad-based and started to show signs of private investment expansion, while the impact from the hurricane on the economy overall was limited. Nevertheless, the passing of the U.S tax reform bill in late December 2017 would partly help boost further economic growth in the future. **Euro area economies continued to grow** driven mainly by consumption and exports. In the period ahead, accommodative financial conditions, robust consumer confidence and gradual recovery of the labor market would support consumption as a key economic growth driver. **Japan's economy was projected to expand** thanks to the expansion of exports and manufacturing in tandem with global trade recovery. In addition, private consumption would continue to expand on the back of robust consumer confidence and the improved labor market.

Looking ahead, growth of advanced economies could face risks stemming from (1) uncertainties pertaining to U.S. economic policies such as foreign trade policy and infrastructure investment policy, (2) negotiations on trade and other issues between the U.K. and the European Union after Brexit, and (3) political uncertainties in Europe, particularly on government formation in Germany and the election in Italy.

China's growth slightly slowed down. Meanwhile, other Asian economies exhibited better-than-expected growth due to continued improvement in exports supported by global demand and a gradual recovery of domestic demand as confidence of private sector picked up.

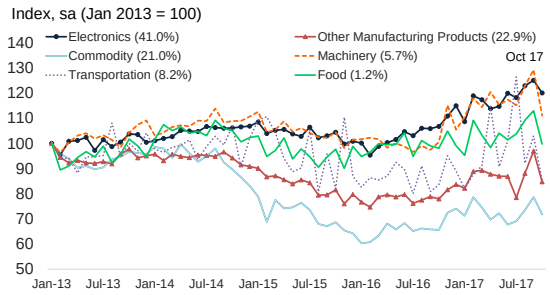
In the third quarter of 2017, China's growth slightly slowed down from the previous quarter. The growth outturn, however, was better than the assessment in the previous *Monetary Policy Report* driven by exports, consumption, and investment in construction and manufacturing. In the period ahead, China's growth would likely slow down due to (1) financial stability measures, where additional measures to manage risks arising from off-balance sheet assets of commercial banks and more stringent lending standards for local administrations were recently announced (anti-speculation measures in the real estate market were already in place since the end of last year) and (2) ongoing economic structural reforms such as a reduction in excess production capacity and a shutdown of factories that caused environmental pollution. Nevertheless, improved global demand and additional targeted financial measures to support SME financing would help sustain a gradual adjustment of the Chinese economy. However, high corporate debt remained an issue for financial system stability risks that warranted close monitoring.

Asian economies (excluding Japan and China) gained further traction driven mainly by exports. Such export expansion was more broad-based across wider range of product categories, including electronics, machinery and equipment, commodities and food

(Chart 1.1). Improved exports also supported manufacturing to gradually recover (Chart 1.2) and boosted corporate performance and confidence. These, in turn, resulted in a stronger employment in some countries, which would help support household consumption in the period ahead. In the third quarter, Asian economies accelerated at a faster pace than usual, partly due to the Hari Raya Aidilfriti holidays which took place in June instead of July and the moon festivals in South Korea and Taiwan which were postponed from September to October. Therefore, there were more working days in the third quarter compared with the previous year and also a temporary acceleration of economic activities. Looking ahead, Asian economies were projected to gradually expand despite some risks from (1) elevated household debt which could weigh on domestic demand recovery in several countries, (2) a faster-than-expected slowdown of the Chinese economy which would undermine overall Asian exports, (3) uncertainties surrounding the U.S. foreign trade policy which could impact global and Asian trade, and (4) geopolitical risks in the Korean peninsula and the Middle East.

Chart 1.1 Overall increasing trend for Asian exports which became more broad-based across product categories

Asian exports value* classified by product categories

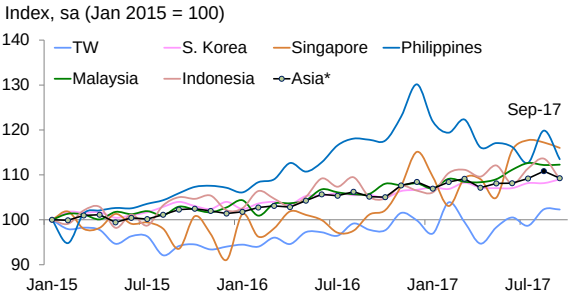


Note: *Asian exports include Hong Kong, Taiwan, S. Korea, Malaysia and Singapore.
( ) share of total exports in 2016
Commodities include crude oil, metals, chemicals, rubber, and vegetable oil.
Other manufacturing products include textile, papers, furniture, footwear and miscellaneous

Source: CEIC

Chart 1.2 Asian industrial production increased in line with exports

Asian industrial production index*



Note: Asia* consists of Indonesia, Taiwan, S. Korea, Malaysia, Singapore and the Philippines weighted by nominal GDP
The latest data is a preliminary data, assuming steady growth for countries whose data is not yet released.

Source: CEIC

The growth outlook for Thailand's trading partners was revised up, with upside and downside risks to the baseline projection largely balanced.

Economic growth of Thailand's trading partners would likely be stronger than the assessment in the previous *Monetary Policy Report*. This was attributable to better-than-expected economic outturns in the third quarter of many countries, coupled with growth momentum in the period ahead from improved economic fundamentals. The Committee thus revised up the growth forecast for Thailand's trading partners to 3.8 percent and 3.5 percent in 2017 and 2018, respectively (Table 1.1).

The Committee assessed that risks to growth of Thailand's trading partners were largely balanced, which improved from the previous assessment that downside risks outweighd upside risks. This was due to higher-than-expected growth in advanced economies and China, which propelled growth of Asian exports. However, there remained risks that warranted monitoring including (1) uncertainties pertaining to U.S. economic policies such as foreign trade policy and public infrastructure investment and (2) geopolitical risks that could intensify and lead to increased volatility in the financial markets, commodity prices, as well as business, trade and tourism confidence.

Table 1.1 Assumption on trading partners' economic growth

Annual change (%YoY)	Weight (%)	2016*	2017	2018
United States	14.9	1.6	2.3 (2.1)	2.5 (2.1)
Euro area	10.0	1.7	2.2 (2.1)	1.8 (1.6)
Japan	13.6	1.0	1.5 (1.5)	1.2 (1.1)
China	15.7	6.7	6.8 (6.7)	6.5 (6.3)
Asia (excluding Japan and China)**	37.4	3.5	4.5 (4.2)	4.1 (3.9)
Total***	100.0	3.1	3.8 (3.6)	3.5 (3.4)

Note: *Outturn

**Weighted by share of exports from Thailand to 7 trading partners in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8%), Taiwan (2.5%), Indonesia (5.9%), South Korea (2.8%), and the Philippines (3.7%)

***Weighted by share of exports from Thailand to 13 trading partners in 2014 (including the United Kingdom and Australia)

( ) as reported in Monetary Policy Report, September 2017

Most central banks maintained their accommodative monetary policy stance. However, some central banks raised their policy interest rate.

Monetary policy of most central banks remained accommodative. However, some central banks raised their policy interest rate such as the Federal Reserve (Fed) and the Bank of England (BOE). The Fed commenced its balance sheet reduction in October 2017 as previously announced and hiked the federal funds rate once in December 2017. Meanwhile, the European Central Bank (ECB) announced its plan to reduce its monthly bond purchase from 60 to 30 billion euros per month from January until September 2018. This was on account of a continued recovery of the euro area economies, while inflation remained subdued. Therefore, accommodative monetary policy was still necessary to support the economies for an extended period. In addition, the Bank of Korea (BOK) raised its policy rate for the first time since 2011 in November thanks to a continuation of a better-than-expected growth of the Korean economy, while inflation was projected to gradually trend up toward the target. Moreover, signs of divergence in monetary policy were also observed among other countries in the region due to a continued economic expansion and rising inflation toward the target, which would allow for monetary policy normalization in the period ahead.

Capital flows in Asia recorded higher net inflows in recent periods following market expectations on the U.S. tax reform policy and the gradual monetary policy normalization of major advanced economies.

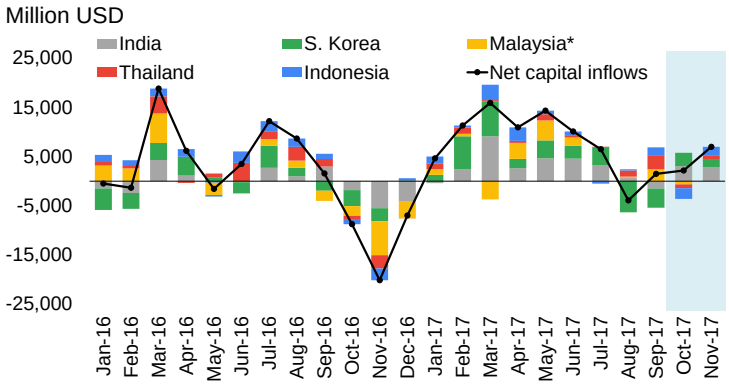
In October 2017, net capital inflows to emerging market Asia (EM Asia) slowed down slightly from the previous period due to uncertainties regarding the appointment of the new Fed chairman, which could affect the U.S. monetary policy stances, and a more progress in a consideration of the draft U.S. tax reform bill. However, capital inflows to EM Asia accelerated somewhat in November 2017 (Chart 1.3) because (1) investors anticipated delay in the U.S. tax reform, (2) the new Fed chairman and central banks of other major advanced economies were poised to gradually normalize their monetary policies, and (3) Asia's economic outlook continued to improve as its economic growth outpaced that of major advanced economies. However, the size of capital inflows varied across countries as investors put more

consideration on country-specific factors including economic fundamentals and the policy rate outlook.

Looking ahead, global financial markets would likely remain volatile. International capital movements might fluctuate, both in and out of Thailand, particularly given monetary policy normalization commenced by several central banks, together with heightened uncertainties on the external front such as the U.S. foreign trade policy and geopolitical risks, whose developments would warrant close monitoring.

Chart 1.3 Capital flows to EM Asia increased from previous quarter due mainly to changing expectations on U.S. economic policies

Capital flows to equity and bond markets in EM Asia*



Note: Foreign portfolio flows to Malaysia's equity market were calculated from international investment position (IIP) which was available until September 2017.

Sources: Institutional Institute of Finance and Bank Negara Malaysia

Crude oil prices rose from temporary factors in the fourth quarter of 2017 but would likely decrease in early 2018 following an increase in supply from shale oil producers in the U.S. Nonetheless, oil prices would slowly trend up in the period ahead in tandem with the global economic recovery.

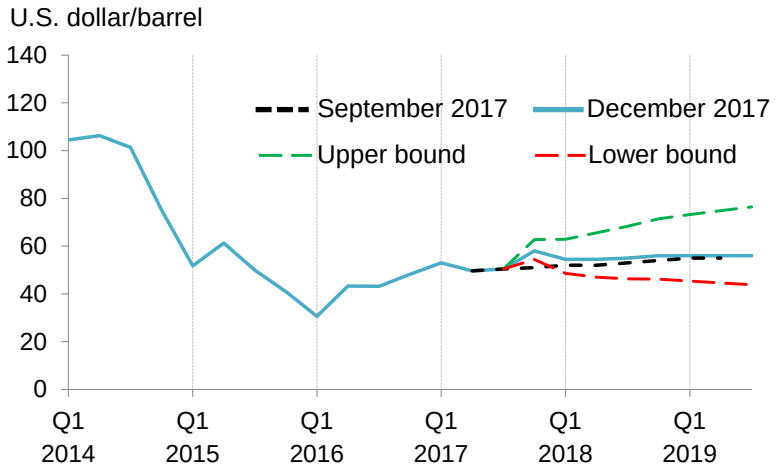
In the fourth quarter of 2017, the Dubai crude oil price increased from the previous quarter. Prices reached a two-year record high in November 2017 due to (1) a gradual decline in oil stock following demand expansion, (2) compliance to the agreement on production cuts among OPEC and Non-OPEC producers, and (3) heightened geopolitical risks in the Middle East. **However, the Committee expected prices in early 2018 to fall** because the oil rig count and oil productions of shale oil producers in the U.S. would likely rise considerably given the continuation of high oil prices. Moreover, geopolitical risks were expected to alleviate. However, crude oil prices would gradually trend up in the period ahead, supported by an extension of the OPEC and Non-OPEC's agreement on production cuts until the end of 2018. This would bring down the excess oil supply to a balanced level, while a pick-up in demand would continue following global economic recovery.

Hence, the Committee revised up the projection for Dubai crude oil prices from 50.9 and 52.8 dollars per barrel to 52.8 and 55.0 dollars per barrel in 2017 and 2018, respectively. Moreover, risks to the projection tilted to the upside instead of balanced as in the previous projection. This was a result of geopolitical risks in the Middle East and the Korean Peninsula that could lift oil prices up in some periods. Moreover, a rise in crude oil supply from the U.S. shale oil producers would likely be slower or lower than previously expected. However, other risks could stem from failure to comply with the agreement among

OPEC and Non-OPEC producers, resulting in a decline or a slower-than-expected increase in crude oil prices.

Chart 1.4 Dubai crude oil price hit 2-year record high due to temporary factors but would likely decrease in early 2018 following increase in U.S. shale oil supply

Assumption on Dubai crude oil price



2. The Thai Economy

2.1 Recent Developments

Thailand's economic growth gained further traction in the third quarter of 2017 from the previous quarter with growth outturn close to the previous assessment in the previous *Monetary Policy Report*. Main drivers were strong expansions of merchandise exports and tourism as well as continued domestic demand growth. Meanwhile, the impact from regulations on immigrant workers gradually subsided.

The Thai economy expanded 4.3 percent in the **third quarter of 2017** from the same period last year, an improvement from 3.8 percent in the previous quarter. The key growth driver was an expansion of merchandise exports across almost all product categories and export destinations. In particular, exports of rice, prawns, and electronics expanded in tandem with improved external demand. Exports of services continued to grow despite some slowdown compared with the previous quarter due to a drop in tourists from Europe and Malaysia. Private consumption expanded in all product categories thanks to improvements in non-farm income and demand for consumer credit. While private investment slowed down due to a contraction in the construction sector, investment in machinery and equipment increased across almost all business sectors. Meanwhile, the public sector remained a key growth driver on the back of an acceleration in public consumption. On the other hand, public investment slowed down as the government's water development and management projects and emergency road projects neared completion. In addition, state-owned enterprises investment was mostly ongoing without disbursement in additional large projects. Overall, the Thai economy recorded a 1.1 percent growth, after seasonal adjustment, in the third quarter of 2017, close to the growth rate of the previous quarter.

Thailand's economic growth continued to gain further traction **over the first two months of the fourth quarter**. Merchandise exports were robust in all major export destinations and almost all product categories in tandem with continued improvements in external demand and global trade recovery. In addition, there were positive spillovers from export expansion to all sizes of businesses which resulting in higher labor income in export-related businesses. Service exports recorded stronger growth on account of increased foreign tourists in almost all groups coupled with the low base effect from last year's measures to curb illegal tour operators. Overall private consumption gradually expanded despite some slowdown in October due to temporary factors. Supporting factors were improved non-farm income and tax measures that implemented to stimulate the economy at the end of last year. However, farm income dropped somewhat from the previous quarter as outputs were affected from flooding in some areas and prices of agricultural products declined. Private investment continued to improve while remaining at a low level. Investment in machinery and equipment increased in line with higher imports of capital goods in several categories, e.g. telecommunication, energy, and photography and film equipment. Meanwhile, sales in construction materials dropped due to flooding in some areas. Public spending increased from both current and investment expenditure. Current spending expanded in line with compensation of employees especially for government pensions. Capital expenditure increased from public investment especially by the Department of Highways, the Department of Rural Roads, and the Department of Irrigation. Meanwhile, state-owned enterprises

investment slowed down following accelerated disbursement for the power plant project and the natural gas pipeline projects previously.

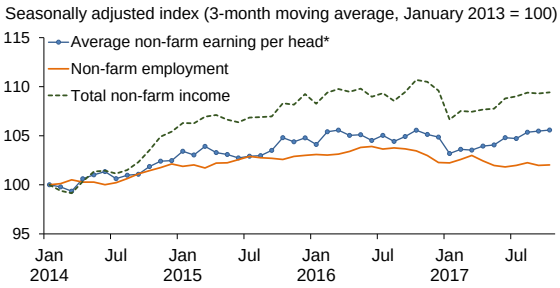
Household purchasing power was not yet sufficiently broad-based. Some agricultural households were affected by declining in agricultural product prices. Meanwhile, nonagricultural households did not fully benefit from the economic recovery, especially the low-income group.

Agricultural household income was higher from the previous year due mainly to higher output, although prices of many agricultural products, e.g., rubber, cattle, and palm oil, declined given higher supply in the market. In addition, some households were also affected by flooding in some areas. **Nonagricultural households** did not fully benefit from the economic recovery. Nonetheless, total non-farm income rose thanks to higher average earnings (wages and overtime payment) per head, particularly for the high-income group whose business benefiting from exports and tourism. Meanwhile, nonagricultural employment declined from the previous quarter (Chart 2.1), especially in the low-income group of daily workers in the manufacturing and construction sectors as well as business owners in the trade sector. While the services sector could absorb labor to some extent (Chart 2.2), some workers might earn less as they moved to sectors with lower productivity and wages.

Lower employment in the manufacturing sector was observed in domestic-oriented businesses, especially for small and medium enterprises (SMEs) that might find difficulty in coping with higher business competition and fast developments of technology. In addition, business models that became less reliant on labor also led to lower employment. For example, e-commerce helped reduce the need to open new branches and hire more employees. Moreover, modern trade businesses could adapt quickly to technology and have lower costs. Nonetheless, the increasing adoption of automation was observed in the manufacturing sector, especially in high-growth export industries such as electronics, automobiles, and plastic. As a result, positive spillovers from export expansion to employment were largely limited. The Committee would thus closely monitor on employment and household income.

Chart 2.1 Overall non-farm income increased in line with average earning per head*, although employment in manufacturing and construction sectors declined

Average non-farm earning per head*, non-farm employment, and total non-farm income

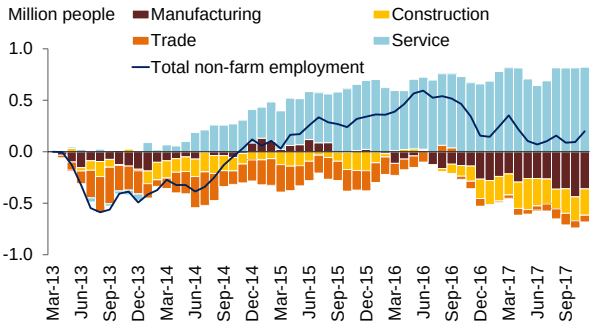


Note: *Average non-farm earning per head was calculated from monthly average wages and overtime payment

Source: National Statistical Office and calculation by Bank of Thailand

Chart 2.2 Service sector absorbed labor migrating from manufacturing and construction sectors

Change in non-farm employment classified by sector, compared to March 2013



Source: National Statistical Office and calculation by Bank of Thailand

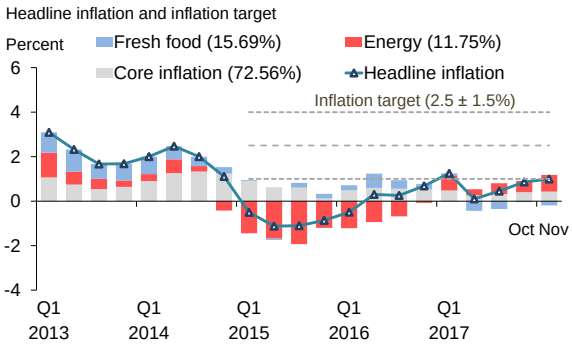
The impact of regulation on immigrant workers subsided but still warranted monitoring.

The government undertook measures to manage and legalize immigrant workers in Thailand through the issuance of the Royal Decree on Managing the Work of Aliens effective on June 23, 2017. Since then, immigrant workers gradually returned after leaving Thailand earlier, and the impact of potential losses of purchasing power of immigrant workers on domestic consumption subsided. Meanwhile, most business owners were able to adjust and continue business activities, although some were faced with higher labor costs, especially for small and medium enterprises in the hotel and restaurant sector, construction, and trade who were highly dependent on immigrant workers. The risk of future labor shortage that might result from immigrant worker registrations and nationality verifications declined after the government stepped up implementation and cooperation with the source countries. However, the Committee would continue to monitor the impacts of these measures.

Headline inflation edged up mainly due to energy prices, while core inflation slightly increased from excise tax restructuring.

Headline inflation averaged at 0.93 percent over the first two months of the fourth quarter, up from 0.45 percent in the previous quarter (Chart 2.3). The increase was attributable to (1) higher domestic retail oil prices in line with global oil prices, (2) higher energy charges (FT) during September – December 2017 following prices of natural gas which constituted a main part of electricity costs, and (3) higher liquefied petroleum gas (LPG) prices in tandem with directions of global LPG prices. Meanwhile, a decline in fresh food prices was less pronounced as last year's high base effects following the drought dissipated. However, this year's weather conditions that were favorable for the output of vegetable and fruits remained a key factor behind a gradual rise in fresh food prices. The one-year-ahead inflation expectations of businesses were at 2.1 percent in November, close to the rate in the previous quarter's survey. Meanwhile, five-year-ahead inflation expectations of professional forecasters were at 1.8 percent in October 2017, a drop from 2.3 percent from the previous survey in April 2017.

Chart 2.3 Headline inflation picked up slightly from previous quarter mainly due to energy prices



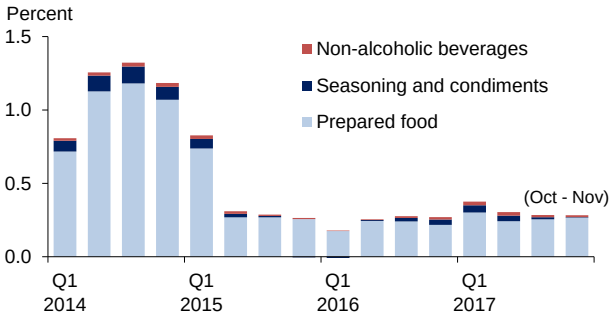
Note: ( ) denotes share in inflation baskets
Source: Ministry of Commerce and calculation by Bank of Thailand

Core inflation averaged at 0.60 percent over the first two months of the fourth quarter in 2017, up from 0.49 percent in the previous quarter. Prices of food items in the core inflation basket were stable at low levels as costs of fresh food and liquefied petroleum gas (LPG) remained low despite some slight increases (Chart 2.4). Prices of non-food components in core inflation slightly rose due to an increase in excise tax on tobacco and alcoholic beverages. Meanwhile, prices of other items slowly trended up following a gradual domestic demand expansion (Chart 2.5). Moreover, structural factors partly put downward pressure on prices of non-food components in core inflation. These factors included (1) globalization that enhanced price competitions and enabled businesses to gain easier access to cheaper raw materials, (2) technological advancements that led to lower costs of production, and (3) the greater role

of e-commerce that led to lower costs and intensified price competitions as consumers could easily make price comparisons.

Chart 2.4 Prices of food items in core inflation remained at low levels, albeit picking up somewhat, as LPG and fresh food prices remained subdued

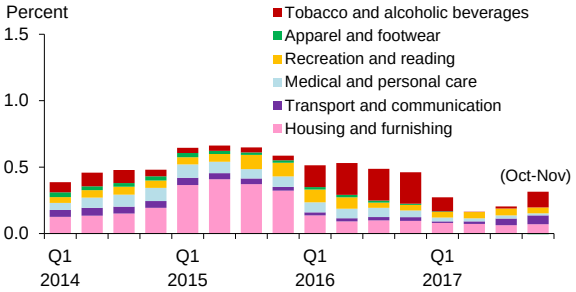
Contribution to food-in-core inflation (28.17%)



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculation by Bank of Thailand

Chart 2.5 Prices of nonfood item in core inflation picked up due to increase in excise tax on tobacco and alcoholic beverages

Contribution to non food-in-core inflation (72%)



Note: *Contribution to inflation displayed composition of changes to inflation weighted by share of each component in inflation baskets

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Table 2.1 Inflation

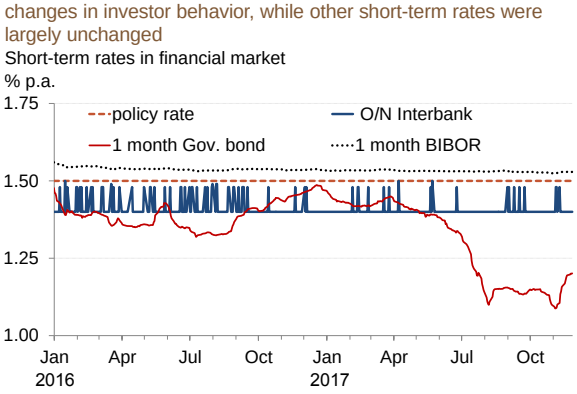
Annual percentage change				2017			
	Q2	Q3	Q4	Q1	Q2	Q3	Oct-Nov
Headline Consumer Price Index (Headline CPI)	0.30	0.26	0.69	1.25	0.10	0.45	0.93
Core Consumer Price Index (Core CPI)	0.79	0.76	0.73	0.66	0.47	0.49	0.60
Raw food	4.24	2.58	1.54	0.61	-2.99	-2.25	-0.74
Energy	-8.95	-7.00	-1.06	6.69	2.67	4.86	5.85

Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Short-term money market rates stayed low, while long-term government bond yields rose from both supply-side and external factors.

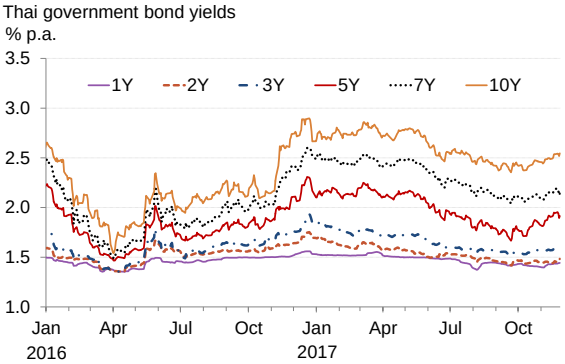
Short-term money market rates remained close to the policy interest rate in the fourth quarter of 2017, except for **short-term government bond yields** that remained below the policy interest rate. This was attributable to both the reduction in the short-term bond issuances by the Bank of Thailand and treasury bills issuances by the Ministry of Finance. However, short-term bond yields edged up somewhat in November as investors turned to long-term bonds which offered higher returns (Chart 2.6). Meanwhile**, medium-term and long-term government bond yields** were volatile in October due to external factors, especially uncertainties pertaining to U.S. tax policy reform. After that, yields gradually pick up owing to the higher supply of Thai government bonds and the U.S. tax policy reform that became more evident and could lead to budget deficits. Consequently, U.S. government bond yields were expected to rise and could lead to higher Thai government bond yields. (Chart 2.7).

Chart 2.6 Short-term government bond yields started to rise due to changes in investor behavior, while other short-term rates were largely unchanged



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

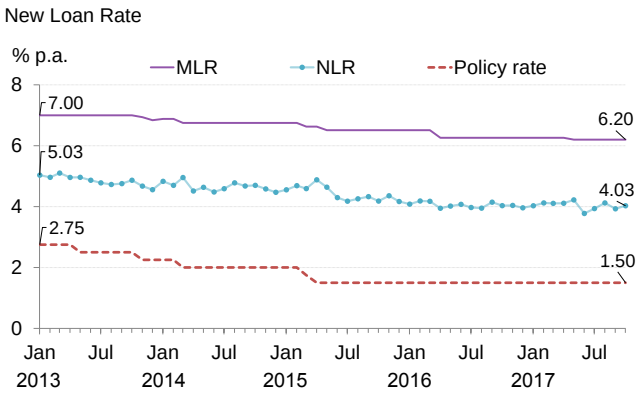
Chart 2.7 Medium- and long-term government bond yields rose owing to domestic supply-side and external factors



Source: Thai Bond Market Association (Thai BMA)

Corporate bond yields edged up alongside increases in government bond yields, particularly for corporate bonds with low credit rating and unrated bonds. Meanwhile, **financing costs through commercial banks**, as reflected in the new loan rate (NLR[1/]), were stable at a low level after having declined continuously (Chart 2.8).

Chart 2.8 New Loan Rate (NLR) was stable at low level after continuously trending down



Source: Bank of Thailand

Overall private credit continued to expand primarily due to acceleration in credit to households. Loans extended to SMEs began to pick up in several businesses, reflecting a more broad-based economic recovery.

Private credit expanded[2/] 3.3 and 3.2 percent in the third quarter and October 2017, respectively, from the same period last year (Chart 2.9). This was largely due to household credit growth, particularly mortgage and auto loans which accelerated after the cars bought under the first-car scheme reached the five-year contract period. On the other hand, overall business credit growth slowed down, especially for credits to businesses in manufacturing and trade sectors. This was partly owing to debt repayment of large corporates. However, credits to SMEs in the food and beverage, real estate and utilities sectors expanded. Working capital

[1/] NLR is calculated based on a weighted average of interest rates for new loan contracts extended by 14 Thai commercial banks (excluding consumer loans and loans to financial intermediaries). The data covers loans of value of 20 million baht or higher for all purposes and terms, and includes both secured and non-secured loans. Moreover, interest rates used in the calculation refer to the mid-rate between the lowest and the highest rates in each loan contract.

[2/] Outstanding credit of other depositary corporations (ODCs), namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and market mutual funds.

loans to export-oriented sectors continued to expand such as electronics, hard-disk drives, and rubber and plastic products.

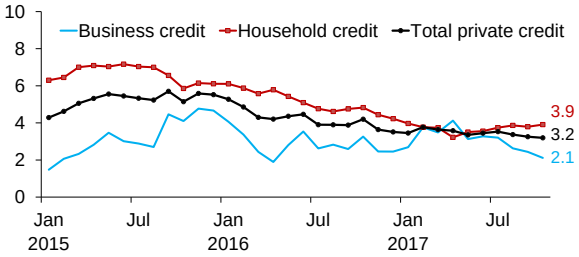
The issuance of corporate bonds expanded at 14.2 percent in the third quarter of 2017 from the same period last year, especially in finance and banking sectors. As of the latest data in October 2017, net issuance of corporate bonds slightly edged down to 13.7 percent with the new issuances dominated by the finance and banking sectors (Chart 2.10). **Funding through the equity market** increased markedly at the beginning of the third quarter of 2017 in businesses related to food, banking, and petrochemical and chemical products. The increase in funding was mainly for financial restructuring and business expansion purposes. Funding through equity continued to rise from both capital increases by several businesses and initial public offering (IPO) by construction material companies to finance development in business operation and efficiency as well as business expansions both overseas and domestically.

Going forward, financial conditions were expected to remain accommodative, as reflected in the real policy interest rate which remained at a low level and was moderate compared with other countries (Chart 2.11). Meanwhile, financing costs through commercial banks, as reflected in the new loan rate (NLR), were stable at a low level. Nonetheless, the Credit Condition Survey[3/] indicated that financial institutions would still maintain caution in extending credit to households and certain businesses, especially SMEs with deteriorating credit quality.

Chart 2.9 Private credit expanded mainly on account of household credit

Growth of private credit

Percentage change from the same period last year



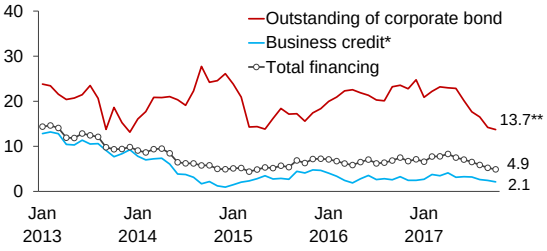
Note: Private credit includes credit to other depositary corporations (ODCs) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds

Source: Bank of Thailand

Chart 2.10 Total financing continued but at slightly slower pace

Growth of corporate bond outstanding and business credit

Percentage change from the same period last year



Note: * Business credit covers lending activities of Other Depository Corporations (ODCs) namely commercial banks, special financial institutions, saving cooperatives and money market mutual funds

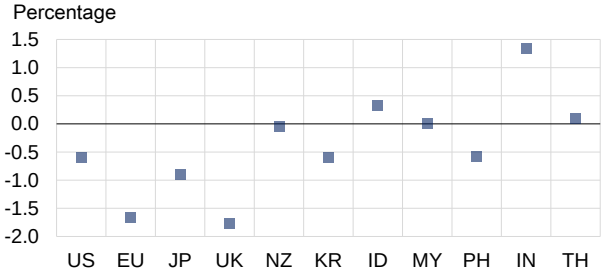
** In August, telecommunication and energy businesses did not roll-over their matured debt during that month, resulting in slower growth of corporate bond outstanding.

Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Chart 2.11 Thailand's real policy rate remained low

Real policy rates*

Percentage



Note: *Calculated from policy rate less 1-year ahead inflation expectation surveyed by the Consensus Forecasts (data as of 4 December 2017)

Source: Consensus Economics and calculation by Bank of Thailand

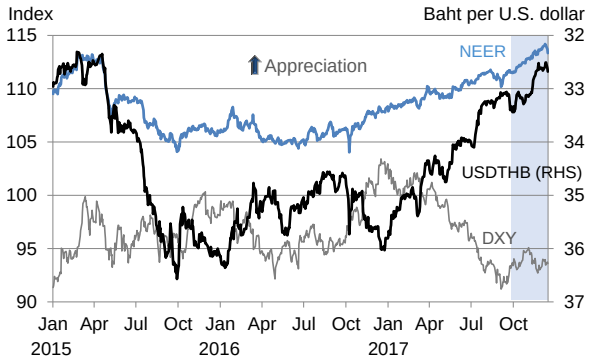
[3/] Report on Credit Conditions Q3/2017 and Outlook for Q4/2017

The baht remained largely unchanged. However, the baht marginally appreciated against the U.S. dollar as the dollar weakened. Meanwhile, the real effective exchange rate appreciated somewhat in line with improvements in economic fundamentals.

In the fourth quarter of 2017, **the baht slightly strengthened against the U.S. dollar** relative to the end of the previous quarter. This was mainly due to the weakening of the U.S. dollar (Chart 2.12), underpinned by negative market sentiments on the U.S. dollar, particularly in mid-November when the U.S. tax policy reform was expected to be delayed. Moreover, expectation of the Fed's gradual rate hikes by the new Fed chairman put downward pressures on the dollar. The ECB's further monetary policy easing and political uncertainty in Europe also played a part in attracting investors to redirect their attention toward the Asian markets, especially in economies with positive economic outlook such as South Korea, Malaysia, and Thailand. Hence, the baht appreciated against major currencies, in line with directions of most regional currencies. Despite the Fed's rate hike[4/] and the on-track progress of the U.S. tax policy reform, the exchange rates were broadly unchanged. However, during the latter half of December, the baht weakened against the U.S. dollar partly due to merging and acquisition investment overseas by large corporates. Overall, the baht remained largely unchanged against the U.S. dollar from the previous quarter. As of December 19, 2017 the baht appreciated 2 percent from the end of the previous quarter, closing at 32.68 to the dollar.

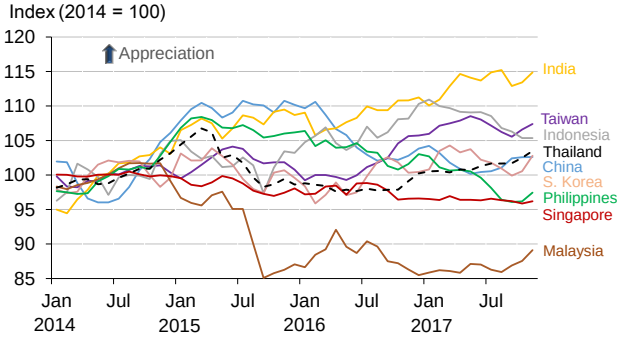
The **nominal effective exchange rate (NEER)** stood at 113.34 on December 19, 2017 which represented a 1.6 percent appreciation from the end of the previous quarter, driven by the baht's appreciation against major currencies, namely the Japanese yen, the U.S. dollar, the Chinese yuan, and the Australian dollar. As of the end of November 2017, the **real effective exchange rate (REER)** rose by 4 percent from the end of last year, which was less than the NEER appreciation of roughly 5 percent. This was owing to Thailand's relatively low inflation compared with trading partners which helped alleviate to some extent the impact of the NEER appreciation on price competitiveness. In addition, Thailand's REER appreciated somewhat consistent with improvements in economic fundamentals and was broadly in line with these of other currencies (Chart 2.13). In the period ahead, exchange rates would likely remain volatile due to external factors including uncertainties surrounding U.S. economic policies, monetary policy directions of major advanced economies, and geopolitical risks.

Chart 2.12 Baht strengthened against weak U.S. dollar
USDTHB, NEER, DXY



Sources: Bank of Thailand and Reuters (data as of 19 December 2017)

Chart 2.13 Thailand's real effective exchange rate appreciated only slightly in comparison to those of other regional economies
Real effective exchange rate (REER) by country



Source: BIS and calculation by Bank of Thailand (data as of December 2017)

[4/] On December 13, 2017, the Fed raised its federal fund rate by 25 basis points to 1.25-1.50 percent with a non-unanimous vote. The markets therefore anticipated that the Fed's future rate hikes could be slower than expected.

Financial stability remained sound overall. However, there remained pockets of risks that warranted monitoring, including (1) debt serviceability of SMEs and low-income households given the economic recovery which was not yet broad-based, (2) search-for-yield behavior, and (3) oversupply of condominium units in some areas.

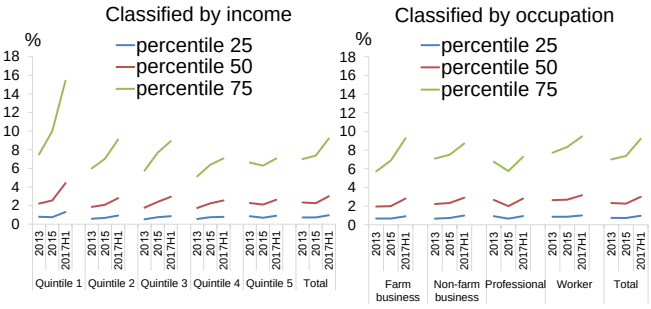
Thailand's financial stability remained sound overall with strong external position as reflected in a high level of foreign exchange reserves and high liquidity of foreign currencies reflecting in sustained current account surplus which would provide cushion against volatilities in global financial markets. At the same time, large corporates showed positive outlooks on the back of stronger financial positions. Financial institutions continued to be sound thanks to high levels of capital buffers and provisions for loan losses to cushion against risks stemming from deteriorating credit quality. Nonetheless, the Committee assessed that there remained pockets of risks that could lead to the buildup of vulnerabilities in the financial system in the period ahead and thus warranted close monitoring. These risks are summarized as follows.

(1) Debt serviceability of both SMEs and low-income households remained fragile as the positive spillovers of economic growth was not yet sufficiently broad-based. SMEs, particularly in manufacturing, trade and construction sectors that operated based on a traditional model, faced difficulties in business adjustments and cost managements. As a result, these firms had fragile financial positions, reflecting in deterioration in the operating profit margin, the interest coverage ratio, and credit quality. The ratio of nonperforming loans (NPL) among SMEs stood at 4.6 percent in the third quarter of 2017, up from 4.4 percent in the previous quarter. However, credit quality of export-oriented SMEs showed signs of improvement. Default risks of corporate bonds did not significantly increase among both rated and unrated bonds as the new issuance of unrated bonds continued to fall. The Committee would closely monitor corporate bonds that would mature in the period ahead, especially those issued by businesses with fragile financial positions.

Debt serviceability of households deteriorated, as reflected in the rising NPL ratio in the third quarter of 2017, particularly mortgage loans. Moreover, the persistent low interest rates had some impact on households' saving behavior. The ratio of household debt to financial assets (proxied for debt accumulation relative to savings) continued to rise for all income groups. This was particularly the case for low-income households as they had limited cushions against economic volatilities (Chart 2.14).

Chart 2.14 Household financial cushion deteriorated

Debt to financial assets ratio



Note: Calculated based on indebted households.

1/ Quintile 1 has the lowest income/head/year. Quintile 5 has the highest income/head/year.

2/ Professional households include managers, academics and professionals, technicians

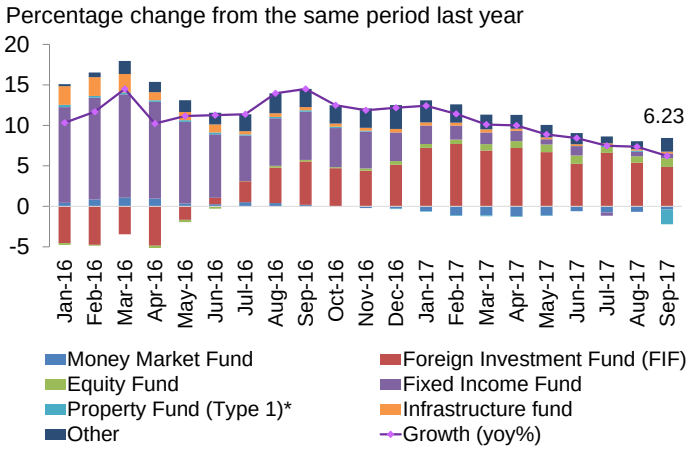
3/ Worker households refer to workers in agriculture, forestry, fishery, machine operations, clerks, services, handicrafts, production processing

Source: Socio-Economic Survey, National Statistical Office, and calculations by Bank of Thailand

(2) The search-for-yield behavior and any signs of underpricing of risks must be monitored as interest rates in major advanced economies began to rise. The prolonged period of low interest rates in Thailand remained a key factor in prompting Thai investors to search for higher return by investing in riskier assets. Despite limited systematic risks overall, signs of underpricing of risks still warranted monitoring. In particular, foreign investment funds

(FIF) continued to expand, with increasing concentration risk (Chart 2.15). Although most funds invested in deposits and short-term debt securities in countries with investment-grade credit ratings, there was still risk of investment concentration in some countries. Moreover, savings cooperatives continued to expand their investments in risky assets such as bonds and equity. Thus, the Committee would closely monitor linkages between saving cooperatives and the financial system through commercial banks and specialized financial institutions.

Chart 2.15 FIF investment continued to expand at high rates
Growth of mutual funds and contributions to growth



Note: *Property Fund (Type 1) is a mutual fund that invests in real estate with regular income from rents, e.g. offices, where the profits will be distributed to unit holders in forms of dividends.
Source: Association of Investment Management Companies (AIMC)

(3) Oversupply of condominium units was found in certain areas. Although overall risks were stable, there remained the need to continue monitoring developments in the property market. First, there was a longer "time-to-go" for condominium units along the MRT Purple Line (Khlong Bang Phai–Tao Poon) and the MRT Blue Line (Bang Sue–Tha Phra) compared with the average of condominium units in Bangkok and its vicinity. Second, the impact from investment in mixed-used real estate projects must also be monitored. Mixed-used projects consisted of both residential and commercial uses such as offices and shopping centers in the same area. Over the next 1-3 years, the impact of their launches was not yet a concern, as most large-scale projects are currently under construction and demand for each type of real estates could absorb the rising supply. However, when large-scale mixed-used projects are completed in the next 4-5 years, there might be an acceleration of supply of residential, office, and retail projects. Consequently, an oversupply might occur in the case where the economy does not turn out as expected by businesses, and developers are unable to make adjustments regarding their launches according to the real estate market cycle. As a result, financial institutions that have lent to these developers would be affected because these borrowers rely on funding through commercial banks as their main financing channel.

2.2 Outlook for the Thai economy

Under the Committee's assessment, Thailand's economic growth was projected to gain further traction going forward and attain a higher growth rate of 3.9 percent in both 2017 and 2018, compared with the assessment in the previous *Monetary Policy Report*. The key growth drivers included (1) robust expansion of merchandise exports and tourism in line with a stronger growth outlook of Thailand's trading partners, (2) a gradual rise in private consumption that began to be more broad-based, and (3) a sustained fiscal impetus despite delay in some investment projects. Meanwhile, inflation was projected to remain at a low level due mainly to supply side-factors but would gradually trend up.

Table 2.2 Summary of forecasts

Percent	2016*	2017	2018
GDP growth	3.2	3.9 (3.8)	3.9 (3.8)
Headline inflation	0.2	0.7 (0.6)	1.1 (1.2)
Core inflation	0.7	0.6 (0.6)	0.8 (0.9)

Note: * Outturn

( ) Monetary Policy Report, September 2017

Sources: NESDB, Ministry of Commerce, estimations by Bank of Thailand

Summary of the key forecast assumptions

- Trading partner economies were projected to achieve higher growth rates than previously assessed as most growth outturns in the third quarter of 2017 were better than expected. In particular, Asian economies continued to grow on the back of exports. In addition, as China's financial stability risks subsided, the authorities were able to implement additional financial measures aimed at specific groups to support business financing.
- The federal funds rate was raised as expected in the FOMC meeting in December 2017. The Fed was expected to raise the policy rate three times in 2018 and to gradually commence its balance sheet reduction in accordance with the announced plan.
- Asian currencies (excluding the Chinese yuan) were stronger than the previous assessment given the outturns in the third quarter of 2017. In addition, a number of central banks in Asia started to signal changes in their monetary policy direction, as reflected in a gradual rise in monetary policy normalization that was sooner than expected.
- The Dubai crude oil price was projected to rise following the extension to the production cut between OPEC and non-OPEC from March 2018 to the end of 2018.
- Farm income was revised down from the previous assessment due mainly to lower agricultural prices resulting from higher-than-expected agricultural output, especially rubber, livestock, and palm oil, thanks to favorable weather conditions.
- Public spending at current prices was revised down from the previous assessment. This was due to (1) a lower-than-expected disbursement of investments by central government owing to limited disbursement efficiency, a reduction in the additional budget for 2018, and the holdover of some state-owned enterprise investment projects from 2017 to 2018, and (2) a downward revision of public consumption as outturns were lower than expected in the third quarter of 2017.

Table: Summary of forecast assumptions

Annual percentage change	2016*	2017	2018
Dubai crude oil price (U.S. dollar per barrel)	41.4	52.8 (50.9)	55.0 (52.8)
Farm income (% YoY)	1.4	4.4 (6.3)	4.1 (4.3)
Government consumption at current price (billion baht)[1/]	2,456	2,552 (2,566)	2,708 (2,710)
Public investment at current price (billion baht)[1/]	936	953 (987)	1,067 (1,123)
Fed funds rate (% at year end)	0.63	1.38 (1.38)	2.13 (2.13)
Trading partners' GDP growth (% YoY)[2/]	3.1	3.8 (3.6)	3.5 (3.4)
Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)3/	154.8	155.8 (156.3)	154.3 (156.4)

Notes: [1/] Assumption includes spending on infrastructure investment plans
[2/] Weighted by each trading partner's share in Thailand total exports
[3/] Increasing index represents depreciation, decreasing index represents appreciation
* Outturns
( ) Monetary Policy Report September 2017

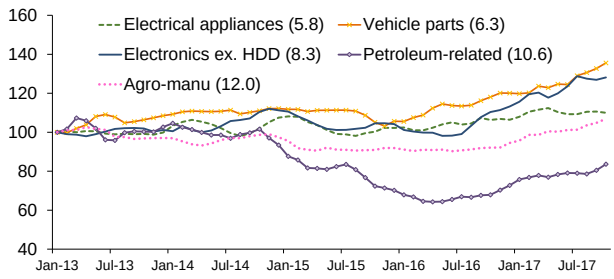
The expansion in merchandise exports was broad-based across product categories and export destinations.

The value of merchandise exports was expected to record higher growth than those reported in the previous *Monetary Policy Report* and register 9.3 percent and 4.0 percent in 2017 and 2018, respectively. Thai merchandise exports gained further traction across various product categories, particularly electronics, auto parts, and processed agricultural products (Chart 2.16), and across almost all export destinations thanks to continued expansion of the global economy and global trade volume (Chart 2.17). Moreover, the export of electronics were expected to record higher growth, especially integrated circuits which would be used in Internet of Things (IoT) devices and electronic parts in automobiles. Exports of smartphones and motorcycles expanded in line with the relocation of production bases to Thailand. In addition, export prices were projected to rise in line with crude oil prices, especially for commodities and oil-related exports such as petroleum products.

Chart 2.16 Merchandise exports continued to expand across several product categories

Value of Thai exports classified by product categories

Seasonally adjusted index, 3-month moving average
(January 2013 = 100)

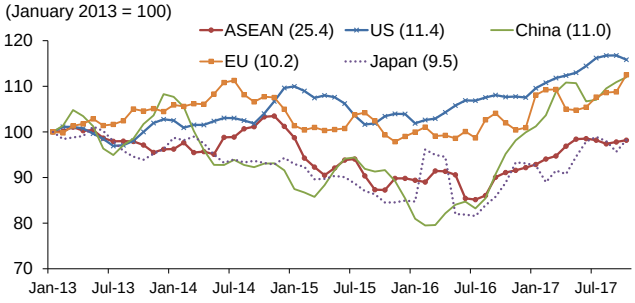


Note: Number in ( ) denotes share of total exports in 2016.
Source: Customs Department and calculations by Bank of Thailand

Chart 2.17 Merchandise exports expanded across almost all export destinations

Value of Thai Exports Classified by Destination

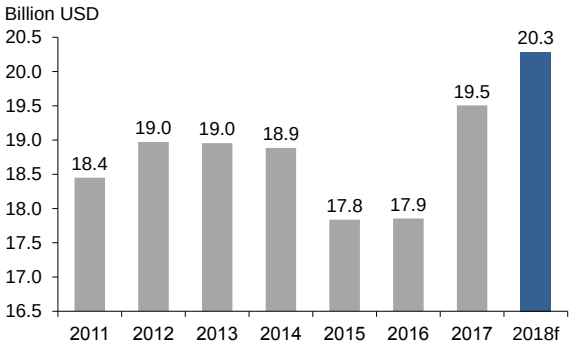
Seasonally adjusted index, 3-month moving average
(January 2013 = 100)



Note: Number in ( ) denotes share of total exports in 2016
Source: Customs Department and calculation by Bank of Thailand

However, the growth outlook of Thai exports in 2018 was projected to expand at a slower pace both in terms of volume and price due to several reasons. First, global trade volume growth and commodity prices were expected to slow down after having accelerated in 2017. Second, there was one-time effect of the relocation of production bases to Thailand, such as automobile tires, hard disk drives, and smart phones. Third, export growth was expected to slow down in some products such as (1) rubber exports to China given a high level of rubber stocks and (2) rice exports after accelerating earlier as several countries experienced natural disasters. There were also structural problems in certain industries that might take some time to be resolved. Nonetheless, the monthly average value of merchandise exports was projected to reach 20.3 billion U.S. dollar in 2018 that would be a record high in many years (Chart 2.18). However, export growth could expand at the rate exceeding the base projection rate as the trading partners' economies might exhibit stronger-than-expected growth on the back of U.S. economic stimulus policies and higher-than-expected oil prices. On the other hand, risks to the export outlook included uncertainties pertaining to U.S. foreign trade policy, geopolitical risks that could undermine the global economic outlook, and a potential slowdown in China as a result of ongoing economic reforms.

Chart 2.18 Merchandise export value was expected to remain high in 2018
Value of exported goods (monthly average)



Source: Customs Department, Ministry of Commerce and calculation by Bank of Thailand

Export of services was revised down slightly in 2017 but was projected to gain further traction in 2018 supported by improvements in the tourism sector.

Export of services was revised down slightly in 2017 as outturns for non-tourism receipts were lower than expected. Export of services was projected to expand at a rate close to the previous assessment in 2018 on the back of robust growth of the tourism sector. Accordingly, the Committee maintained the projection for the number of foreign tourists at 35.6 and 37.3 million in 2017 and 2018, respectively. Factors supporting the tourism sector are (1) an increasing number of Chinese tourists, both group tourists with higher spending per head and independent tourists with high purchasing power, partly thanks to the opening of new direct flight routes from China to major tourist destinations in Thailand, (2) a rising number of ASEAN tourists which was in line with regional economic growth, (3) the reduction and exemption of tourist visa fees which yield benefit to increasing numbers of foreign tourists in the previous period (the visa exemption now ceased to apply but this had only marginal effects on the overall number of tourists), and (4) high growth of tourism receipts that was in tandem with global economic expansion as well as higher-spending tourists.

Given improvements in the value of merchandise and services exports, the projection for the value of merchandise and services imports was revised up with higher imports of raw materials and intermediate goods and higher oil prices. Consequently, **the current account would continue to record large surplus and would be higher than previously estimated, registering surplus of 48.6 and 43.1 billion U.S. dollars in 2017 and 2018, respectively**.

Private consumption was projected to expand at a gradual pace.

Private consumption was projected to gradually expand going forward supported several reasons: (1) improvements in farm income on account of higher agricultural output, despite falls in prices of some products, (2) improvements in non-farm income especially in the export-oriented manufacturing and tourism sectors, (3) household debt deleveraging from completion of debt repayment on the first-car scheme that reached the five-year contract period, and (4) benefits from the government policies, such as the social welfare card project to support low-income individuals, the 9101 project, and flood relief measures[5/]. Nevertheless, employment and income had yet to fully benefit from the overall economic growth, partly due to economic structural changes that in effect brought about less reliance on labor, e.g, automation in industries with strong exports growth and migration of workers from the manufacturing sector to the services sector which has lower productivity and lower wages. In addition, elevated household debt, especially in low-income households, remained a drag on consumption going forward. As households would need some time to shore up their financial position, private consumption was projected to gradually increase and might not be sufficiently broad-based.

Public spending remained a key growth driver despite delay in some investment projects.

Public spending remained a key growth driver. Both public consumption and investment expenditure continued to expand despite delay in central government investment projects toward the end of 2017 which were constrained by limited disbursement efficiency of some agencies and heavy rain which affected construction. Investment projects of state-owned enterprises (SOEs) were mostly on track, although some projects were delayed. These included Airport of Thailand Plc (AOT)'s Suvarnabhumi Airport development project phase 2 which was under the standard price review, the Mass Rapid Transit Authority (MRT)'s Purple Line project (Tao Poon–Rasburana) which resulted in a delayed completion of the terms of references, and Electricity Generation Authority of Thailand (EGAT)'s upgrade and expansion of the power transmission system phase 12 which had problems over site access. Nevertheless, the overall government investment expenditure framework was revised up for 2018 and the majority of the budget was allocated to agencies with high disbursement efficiency. Going forward, state-owned enterprise investment would likely trend upward on account of the holdover of AOT, MRT, and EGAT's investment projects from 2017.

In addition, the promulgation of the Public Procurement and Supplies Management Act, B.E. 2560[6/] might result in a delayed disbursement of some state agencies that were not accustomed to the new system, such as local administrative organizations, while central government agencies which were already accustomed to such system would be less affected by the Act.

[5/] The 9101 project is a project for sustainable agricultural development in honor of His Majesty the late King.

[6/] The Public Procurement and Supplies Management Act, B.E. 2560 was an upgrade of the Regulations of the Office of the Prime Minister on Procurement, B.E. 2535 and the Regulations of the Prime Minister on Electronic Procurement, B.E. 2549 to be an Act. The Act enforces all state agencies to come under the same standards in order to increase efficiency and transparency of procurement process.

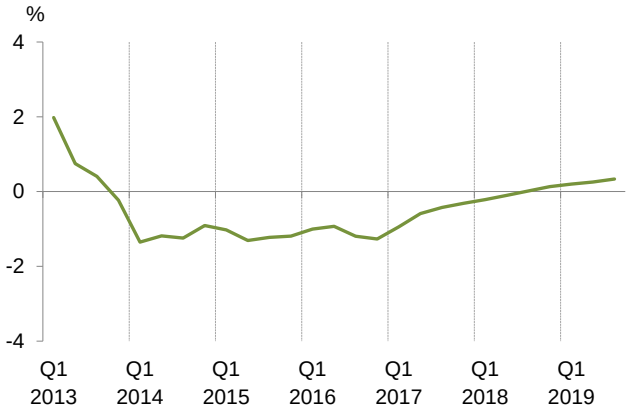
Private investment was projected to gradually recover.

Private investment was projected to recover albeit at a slower pace. Increased imports of capital goods and reduced excess capacity were observed in several industries in tandem with a stronger expansion of private consumption and exports. However, some businesses deferred investment due to excess production capacity. Going forward, private investment was projected to expand in line with continued expansion of exports and private consumption. This was partly reflected in the number of applications for Board of Investment (BOI) investment privileges that was expected to be higher than the previous year[7/]. Moreover, government policies were expected to play an important role in supporting private investment, especially public investment in infrastructure projects and development projects under the Eastern Economic Corridor (EEC). These measures would help shore up business confidence and attract greater foreign investment (Box: Crowding in of private investment by public investment).

Inflation stabilized at a low level due to supply-side factors but was expected to slowly rise going forward.

In the recent periods, inflation remained stable at a low level close to the previous estimate. This was due to a gradual expansion of domestic demand coupled with a fall in fresh food prices due to higher agricultural outputs thanks to favorable weather conditions.

Chart 2.19 Output Gap



In the period ahead, inflation was projected to slowly edge up as **demand-pull pressures** would gradually rise in tandem with the stronger growth outlook. This was reflected in the closing of the output gap in the latter half of 2018 (Chart 2.19). However, demand-pull pressures would be constrained by economic growth that was not yet broad-based and continuous sales promotion offered by businesses. **Cost-push pressures** were expected to rise at a slower pace in 2018 compared with the latter half of 2017 given acceleration in oil prices in the previous period. Cost-push pressures were expected to rise in tandem with an increase in excise taxes on liquor, beer, tobacco and sugary beverages. However, there remained pressures that would cause inflation to rise at a slower pace. These included fresh food prices that were expected to remain low and structural changes following technological advancements and higher price competitions. The Committee therefore projected headline inflation at 0.7 and 1.1 percent in 2017 and 2018, respectively. Core inflation was projected at 0.6 and 0.8 percent in 2017 and 2018, respectively. The Committee assessed that headline inflation would return to the mid-point of the target in the first half of 2018.

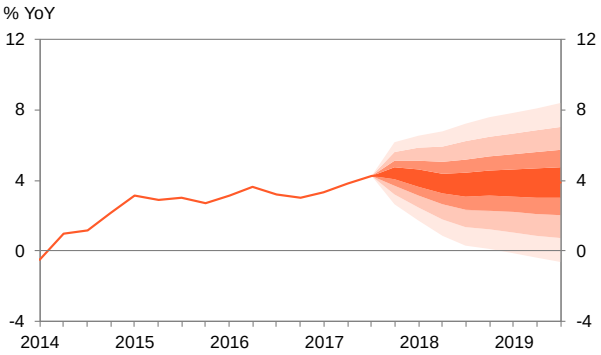
[7/] The BOI requires promoted companies to begin operations within three years. However, more than 60 percent of promoted projects actually undertook investments within 1.5 years.

Risks to the growth and inflation projection were expected to be balanced.

Under the Committee's assessment, the risks to the growth forecast became balanced compared with the downward bias in the previous assessment (Chart 2.20). On the upside, there were possibilities that growth could outperform the baseline projection given a better growth outlook of Thailand's trading partners on the back of the passage of the U.S. tax reform law, additional financial measures to shore up China's economic growth, and higher-than-expected export growth of Asian countries. On the downside, there were risks to the baseline projection due to uncertainties pertaining to U.S. foreign trade policy and geopolitical risks, which might undermine Thailand's trading partners, and lower-than-expected domestic spending as improvement in purchasing power was not yet sufficiently broad-based.

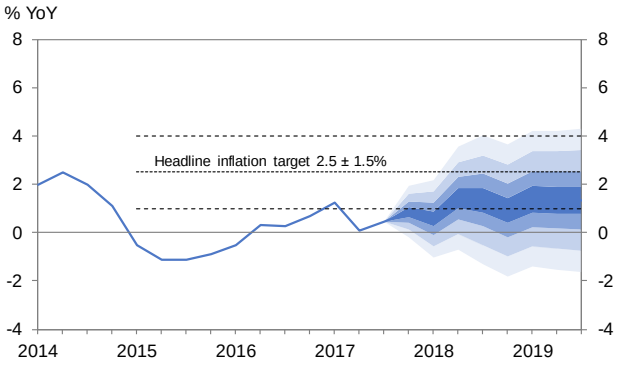
With regard to inflation, the Committee assessed the risks to headline and core inflation to be in balance (Charts 2.21 and 2.22). On the upside, inflation might exceed the baseline projection as crude oil prices could rise on the back of continued global economic expansion and heightened geopolitical risks, lower-than-expected shale oil production in the U.S., and a possibility of minimum wage increases in 2018. On the downside, inflation might fall below the baseline projection as demand-pull pressures could be weaker than anticipated as improvement in purchasing power was not yet sufficiently broad-based.

Chart 2.20 Growth forecast



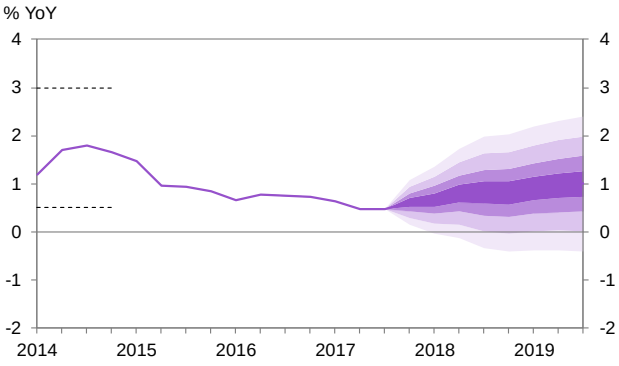
Note: Fan chart covers 90% of the probability distribution

Chart 2.21 Headline inflation forecast



Note: Fan chart covers 90% of the probability distribution

Chart 2.22 Core inflation forecast



Note: Fan chart covers 90% of the probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2016*	2017	2018
GDP growth	3.2	3.9 (3.8)	3.9 (3.8)
Domestic demand	2.7	2.4 (3.0)	3.4 (3.5)
Private consumption	3.1	3.1 (3.3)	3.1 (3.0)
Private investment	0.4	1.6 (2.3)	2.3 (3.0)
Government consumption	1.7	1.5 (2.1)	3.2 (2.7)
Public investment	9.9	0.9 (5.0)	9.0 (9.8)
Exports of goods and services	2.1	6.1 (5.9)	3.7 (3.3)
imports of goods and services	-1.4	6.4 (6.5)	3.5 (3.3)
Current account (billion, U.S. dollars)	48.2	48.6 (42.4)	43.1 (38.6)
Value of merchandise exports	0.1	9.3 (8.0)	4.0 (3.2)
Value of merchandise imports	-5.1	14.0 (14.0)	7.5 (6.3)
Number of foreign tourists (million person)	32.5	35.6 (35.6)	37.3 (37.3)

Note: *Outturns

( ) Monetary Policy Report September 2017

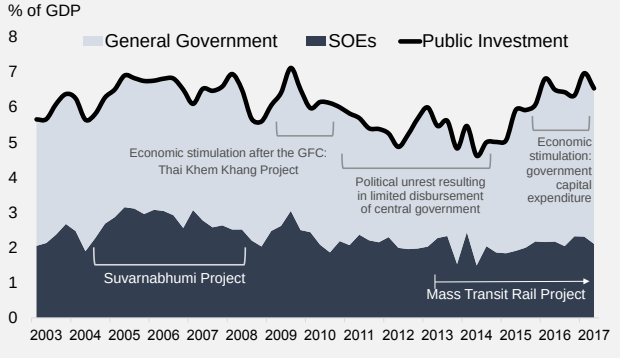
Crowding in of private investment by public investment

Since the global financial crisis in 2008, the Thai economy has been affected by economic contraction in major advanced economies. The contraction was particularly seen in exports and was one of the reasons for the private investment slowdown. The government thus stepped in to play a role in stimulating the economy in the short run and expediting several large-scale investment projects, which would in turn accelerate private investment. Nevertheless, although decision to invest was subject to several factors, such as economic conditions, business competition, and operating costs, **a number of studies suggested that public investment, especially mega infrastructure projects[8/], was one of the key factors having crowding-in effects on private investment**. Such effects could occur both directly and indirectly through several channels including (1) creation of demand for goods and services, (2) cost reduction and increase in competitiveness through infrastructure developments, (3) business opportunities such as development of transportation networks that facilitated urbanization, and (4) improvement in private sector confidence and investment environment.

Large-scale public infrastructure investment helped crowding in private investment.

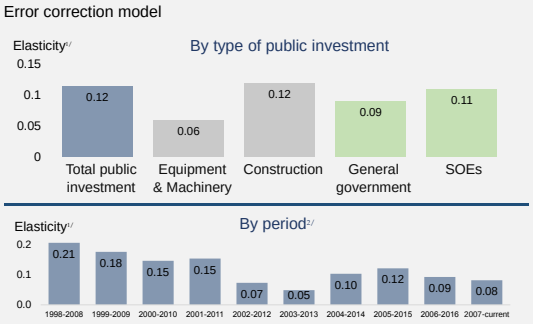
In the past, the Thai government invested in several large-scale public infrastructure projects such as the Eastern Seaboard, Suvarnabhumi Airport, Mass Transit Master Plan 1 and Intercity Motorway (Chart 1). Using data over 20 years, econometric analysis[9/] revealed that a one percent increase in public investment contributed to a 0.12–0.13 percent increase in private investment on average, with the positive effects persisting for 8 quarters[10/] (Charts 2 and 3). Nevertheless, the extent of the increase in private investment depended on types of public investment projects. In particular, construction projects exhibited larger positive spillovers than investment in machinery and equipment. Moreover, a rise in public investment also had positive effects on manufacturing, employment, private sector confidence and overall economic growth.[11/]

Chart 1 Public investment during 2003 - 2017



Note: Data on public investment calculated from System of National Accounts (SNA) and adjusted for price and seasonality factors
Source: Office of the National Economic and Social Development Board, and calculation by Bank of Thailand

Chart 2 Crowding-in effects of public investment on private investment vary depending on types and timing of investment



Note: 1/ Elasticity calculated from change in private investment to change in public investment, where public investment is set to increase by 1 percent over 1 year $(elasticity = \frac{\%\Delta\ Private\ Investment}{\%\Delta\ Public\ Investment})$
2/ calculation based on rolling windows

8/ IMF (2014), "Is it time for an infrastructure push? The macroeconomic effects of public investment", *World Economic Outlook*, October, chapter 3.

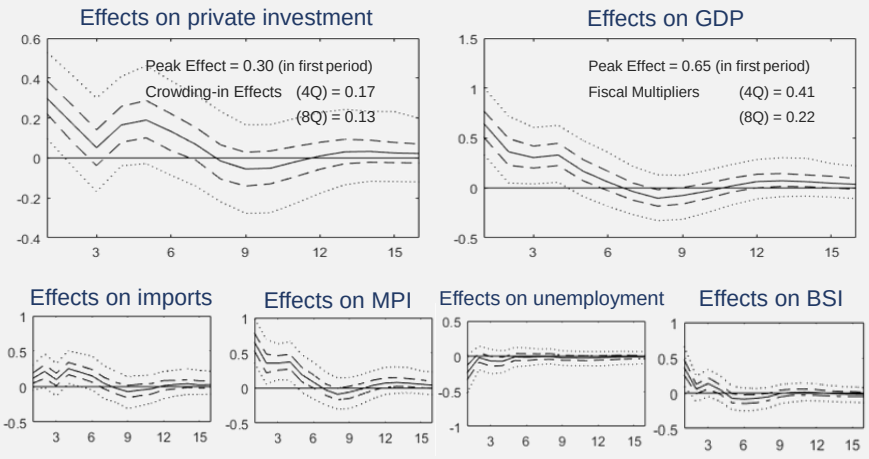
9/ Based on Error correction model and factor-augmented vector autoregression (FAVAR).

10/ Positive effects on private investment as suggested in the FAVAR model were statistically significant during the first 8 quarters after public investment took place. Nonetheless, actual effects might be smaller because the model using a partial equilibrium approach did not take into account the impact of public investment on other economic variables, such as an increase in prices of construction materials and imported raw materials, which could in turn affect private investment.

11/ As reflected in the impulse responses in the FAVAR model of the Manufacturing Production Index (MPI), Business Sentiment Index (BSI), the unemployment rate, and GDP.

Chart 3 Crowding-in effects of public investment on private investment and other variables

Factor-augmented Vector Autoregression (FAVAR)



Note: Impacts of a one percent increase in public investment on private investment and other variables, where the dashed and dotted lines represent confidence intervals of 68 and 90 percent respectively

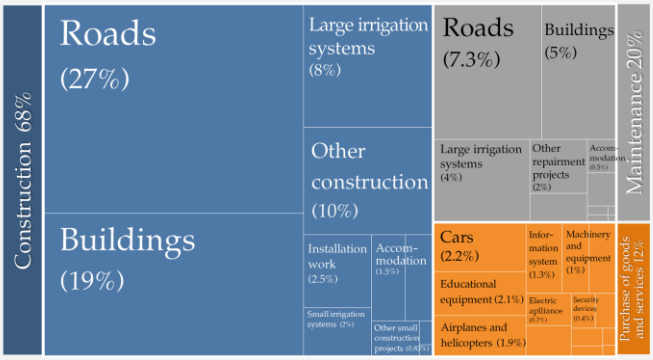
After the global financial crisis in 2008, the crowding-in effects of public investment on private investment were smaller than in the past. This was partly because the government needed to focus on investment in small- and medium-sized projects for which funds could be quickly disbursed in order to stimulate the economy.

Following the global financial crisis, the crowding-in effects of public investment on private investment in Thailand fell by more than half. According to the study, during 2007–present, a one percent increase in public investment led to a 0.08 percent increase in private investment, which was lower than 0.21 percent increase in the previous period during 1998–2007 (Chart 2). **This suggested weaker crowding-in effects of public investment on private investment than in the past,** though such effects were more prominent during certain periods with large-scale investment projects such as the Suvarnabhumi Airport during 2004–2007 and mass transit projects during 2013–2015.

Key factors contributing to the smaller crowding-in effects than in the past were as follows. First, the government had to stimulate the economy in the short run and therefore accelerated investment in small- and medium-sized projects in order to quickly inject money into the economy. Such projects included the Thai Khem Khang project and loans for water resource and road network management projects in 2014. **Second, several large-scale investment projects, which were expected to crowd in private investment, were still at their initial stages,** especially the transport infrastructure investment action plan (priority projects) 2015–2017. **Third, almost 40 percent of the total government capital expenditure during 2014–2017 was spent on improvements of existing projects, purchases of durable goods or equipment, and small-scale investment projects. Crowding-in effects from these expenses were smaller than those from large-scale public infrastructure investment projects** (Chart 4).

Chart 4 Over 40 percent of government capital expenditure was used for construction, purchases of durable articles, and investment in small projects, crowding in only limited private investment

Government capital expenditure for fiscal years 2014-2017 excluding subsidies and central government expenditure



Note: Analysis on budgetary framework based on GFMIS, by project
Sources: Comptroller General's Department and Bureau of the Budget and calculation by Bank of Thailand

However, the findings above only reflected the average effects of public investment on private investment over recent periods. An in-depth measurement of the crowding-in effects would thus require analyses on key investment projects, taking into account linkages with investments in various businesses across different episodes. Some projects might yield higher crowding-in effects than the average. Moreover, upcoming public investment projects would be of different characteristics and under different economic contexts from the past, especially the upgrading of large-scale integrated infrastructure systems such as transportation system and public utility, particularly in the Eastern Economic Corridor area. In such case, crowding-in effects of future public investment were then expected to be larger than the above findings.

The private sector placed emphasis on institutional factors, particularly clarity and continuity of government policy, in making investment decisions.

According to the Business Sentiment Survey by the Bank of Thailand during 2012–2017, **institutional factors such as clarity and continuity of government policy had substantial influences on private sector confidence.** Nevertheless, the private sector wanted the government to place importance on investment in both physical infrastructure, e.g., transportation system, and soft infrastructure, e.g., human capital development, to create labor skills compatible with current state of global development or a more focus on research and innovation development. Not only would such investment help boost confidence and nurture favorable business environment for the private sector in making investment decisions, it could also help raise Thailand's economic potential in the longer run.

In order to increase the crowding-in in the period ahead, the government could consider the following. **First, a focus on necessary infrastructure investment is required in order to create an investment-friendly environment for the private sector in the long run.** This includes development plans for integrated infrastructure systems. **Second, establishing clarity and confidence in government policy is crucial, while also pushing forward investments to continue as planned**. This is particularly the case for the transport infrastructure investment action plan (priority projects), which is the government's policy that is of the private sector's priority and whose investment in construction is expected to have a large positive impact on private investment. **Third, physical infrastructure and soft infrastructure investment must be simultaneously developed,** especially human capital development and innovation. In this way, public investment would raise private sector confidence in making investment decisions. This would help promote sustainable growth of the Thai economy in the long run.

3. Monetary Policy Decision

The Committee deliberated their policy decision by considering benefits and costs of policy alternatives and voted to maintain the current degree of monetary policy accommodation.

In the fourth quarter of 2017, the Thai economy exhibited a stronger growth outlook, particularly on the back of external factors. However, there remained issues to be monitored including the strength of domestic demand recovery, below-target headline inflation, and the buildup of risks to financial stability in certain pockets. Therefore, the Committee had to strike a balance between promoting sustainable economic growth in order to pursue price stability and preserving financial stability. Their conclusions were as follows.

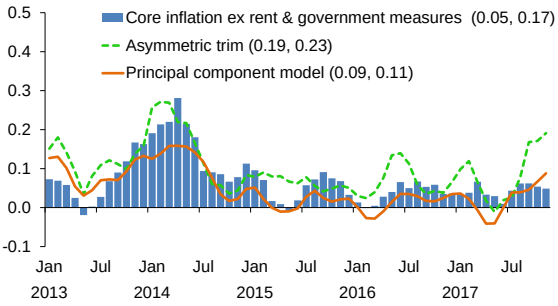
1. Pursuing sustainable economic growth through monetary policy accommodation. The Thai economy was expected to gain further traction driven by continued improvements in merchandise exports and tourism, which were in line with global economic growth and a gradual pickup of domestic demand. Private consumption gradually expanded, as purchasing power and consumer confidence slowly improved. This was because earnings of low-income households, both in agricultural and non-agricultural sectors, did not clearly recover, and household debt remained elevated. Meanwhile, private investment improved as reflected in increases in imports of capital goods and domestic sales of machinery across various industries. Moreover, investment sentiments picked up given greater clarity of the draft Eastern Economic Corridor Act. However, there remained excess production capacity in some industries which might result in a gradual improvement in private investment. Nevertheless, the Committee viewed that monetary policy accommodation partly helped support the continuation of economic growth, although positive spillovers from the economic expansion had yet to sufficiently extend to the labor market and earnings of certain SMEs. This was partly attributable to structural factors that monetary policy alone could not resolve. In addition, the Thai economy would have to face risks from both domestic and external fronts that warranted close monitoring.

2. Returning of inflation toward the target in the medium term. Headline inflation was expected to slowly rise in tandem with domestic demand recovery. This was reflected in inflation indicators that continued to gradually trend up (Chart 3.1). Headline inflation was projected to return to the lower bound of the target within the first half of 2018. Moreover, the Committee viewed that public's long-term inflation expectations that somewhat trended down in recent periods did not reflect deflation risks. This was because inflation was still expected to rise, while prices of most goods and services in the recent period did not decrease (Chart 3.2). In addition, consumption and investment still expanded. However, there were still risks that inflation might return to target slower than projected due to lower-than-expected economic growth, uncertainties pertaining to global oil prices, and impacts of structural changes such as intensified price competition following globalization and e-commerce. Thus, the Committee saw the need to closely monitor and assess changes in inflation dynamics caused by structural factors, as a slower increase in inflation than in the past would significantly impact the pace of the return of inflation to target and the monetary policy conduct going forward.

Chart 3.1 Underlying inflation indicators pointed to higher inflation

Underlying inflation indicators

Percent change from previous month
(3-month moving average, seasonally adjusted)



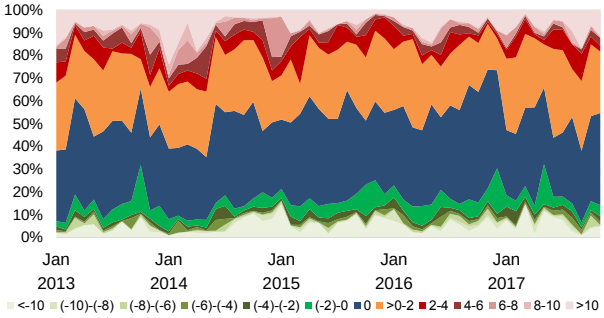
Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of November 2017, while the second value is 2004 - 2014 average; **Asymmetric trim** excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; **Principal component model** calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, and calculation by Bank of Thailand

Chart 3.2 Most prices did not fall, with prices of certain goods revised upward

Contribution to CPI classified by percentage changes of each product.

Percent of CPI



Note: Percent change from the previous month, and annually adjusted (mom%, annualized)

Source: Ministry of Commerce, calculations by Bank of Thailand.

3. Monitoring the buildup of risks to financial stability. The Committee viewed that although financial stability remained sound, but a prolonged low interest rate environment might result in the buildup of vulnerabilities to financial stability in the future. Such vulnerabilities included, in particular, **a continued search-for-yield behavior which could lead to underpricing of risks.** This was reflected in a considerably low volatility in the financial markets. However, if situations did not turn out as expected, market adjustment might increase asset price volatilities. Thus, the Committee would continue to closely monitor developments of significant risks such as the continued expansion of foreign investment funds (FIFs) that was concentrated in only a few countries, although these funds invested in countries with investment-grade credit ratings. Other risks included a substantial increase in asset size and members' deposits in some saving cooperatives whose excess liquidity was invested in risky assets especially when deposit growth outpaced loan growth, as well as investment in non-core businesses by large corporates. Another vulnerability was **debt serviceability of households and SMEs that had yet to improve** as observed in NPLs that remained elevated. Such situation might weigh on their abilities to cushion against economic shocks going forward. Nevertheless, the Committee acknowledged that certain risks could stem from regulatory gaps, and the Committee would thus collaborate between related regulatory authorities in order to undertake tighter measures and stand ready to implement macroprudential measures in an appropriate and timely manner.

The Committee voted unanimously to keep the policy interest rate at 1.5 percent to maintain accommodative financial conditions in order to support a stronger economic growth and foster the return of inflation to target, without causing the buildup of vulnerabilities to financial stability.

The Committee considered benefits and costs of policy alternatives and voted unanimously to maintain the policy interest rate at 1.50 percent at the meetings on 8 November 2017 and 20 December 2017. The Committee saw the need to keep monetary policy accommodative for an extended period in order to support a stronger economic growth and viewed that current level of policy interest rate at 1.50 percent was conducive to

accommodative financial conditions as reflected in a low real policy rate. Moreover, the Committee communicated the need to maintain monetary policy accommodation for an extended period, as it partly helped keep borrowing costs low as seen in bond yields and interest rates charged on new loans that trended down to low levels. With regard to exchange rates in recent periods, the baht appreciated in line with improved economic fundamentals and was moderate relative to other regional currencies. Such appreciation did not significantly affect competitiveness of Thai exporters. However, the baht's movements would likely be volatile due to uncertainties on the external front such as the conduct of monetary and fiscal policies of major advanced economies and geopolitical risks. The Committee would thus continue to closely monitor developments in the foreign exchange market as well as speculative behavior of foreign investors that might affect the baht going forward.

Moreover, the Committee viewed that further monetary policy accommodation would yield low benefits relative to costs, as the effect in boosting a faster return of headline inflation to target would be limited. This was because the recent lower-than-target inflation was due to supply-side and structural factors that could not be resolved directly through monetary policy. On the contrary, further monetary policy accommodation might reduce interests on savings and accelerate the buildup of vulnerabilities to financial stability by way of underpricing of risks and debt accumulation of households. These could in turn affect economic activities.

Looking ahead, the Committee viewed that the degree of current monetary policy accommodation should be maintained for an extended period in order to support a stronger economic growth which could foster a gradual return of headline inflation to target in the medium term. The Committee would closely follow developments and assess the impact of structural factors on inflation dynamics. In addition, the Committee would stand ready to utilize available policy tools to foster the return of inflation to target in a timely manner, as well as to foster the economy to reach its full potential while also preserving financial stability. At the same time, the Committee emphasized the need to develop monitoring processes and assessment of risks to financial stability which could be used in the monetary policy decision-making process going forward.

The Cabinet approved an annual average of headline inflation at 2.5 ± 1.5 percent as monetary policy target for the medium term and for 2018.

On 29 November 2017, the Committee and the Minister of Finance mutually agreed to set an annual average of headline inflation at 2.5 ± 1.5 percent as the monetary policy target for medium term and for 2018. The Cabinet approved the proposed target on 19 December 2017 and viewed that inflation at that level was conducive to the economy to grow in line with potential and would also help maintain competitiveness of the country. Moreover, the tolerance band was appropriate in providing cushion against volatilities that could possibly cause inflation to deviate from the midpoint of the target in the short term. However, given structural changes of both global and Thai economies which would affect inflation dynamics and the inflation outlook in the period ahead, the Committee would closely monitor developments of those changes and of other factors to ensure that the monetary policy target and monetary policy formulation would be more appropriate and effective going forward.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2015	2016	2016		2017		
			Q3	Q4	Q1	Q2	Q3
GDP growth	2.9	3.2	3.2	3.0	3.3	3.8	4.3
Production							
Agriculture	-5.7	0.6	0.9	3.0	5.7	16.1	9.9
Non-agriculture	3.9	3.5	3.2	3.2	3.1	2.8	3.8
Manufacturing	1.5	1.4	1.6	2.2	1.3	1.1	4.3
Construction	17.0	8.3	5.2	6.1	2.8	-6.2	-1.7
Wholesales and retail trade	3.9	5.0	5.2	5.6	5.9	6.0	6.4
Hotels and restaurants	14.6	10.3	13.5	4.9	5.3	7.5	6.7
Transport, storage, and communication	5.1	5.6	6.5	5.2	5.4	8.7	8.1
Financial intermediation	8.8	6.1	5.8	6.7	4.6	5.1	4.8
Real estate, renting, and business activities	1.9	1.8	1.1	1.9	4.0	4.4	4.2
Expenditure							
Domestic demand	2.9	2.7	0.9	2.2	2.3	2.2	2.6
Private consumption	2.2	3.1	3.0	2.5	3.2	3.0	3.1
Private investment	-2.2	0.4	-0.8	-0.4	-1.1	3.2	2.9
Government consumption	3.0	1.7	-5.2	1.8	0.3	2.6	2.8
Public investment	29.3	9.9	5.8	8.6	9.7	-7.0	-2.6
Imports of goods and services	0.0	-1.4	-1.1	3.4	6.1	8.2	6.7
imports of goods	0.2	-2.1	-1.5	3.6	7.3	9.1	8.3
imports of services	-1.0	1.7	0.5	2.0	1.1	4.2	-0.5
Exports of goods and services	0.7	2.1	1.4	1.1	2.7	6.0	7.4
exports of goods	-3.4	0.0	-0.4	1.4	2.6	5.2	8.1
exports of services	17.1	9.3	7.7	0.4	3.2	8.9	4.9
Trade balance (billion, U.S. dollars)	26.8	36.5	9.2	7.1	8.8	6.4	10.1
Current account (billion, U.S. dollars)	32.1	48.2	11.7	10.8	15.0	7.4	13.7
Financial account (billion, U.S. dollars)	-16.8	-21.0	-7.8	-12.1	-7.0	-4.1	1.0
International reserves (billion, U.S. dollars)	156.5	171.9	180.5	171.9	180.9	185.6	199.3
Unemployment rate (%)	1.0	1.0	0.9	1.0	1.2	1.2	1.2
Unemployment rate, seasonally-adjusted (%)	n.a.	n.a.	1.0	1.1	1.1	1.1	1.2

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

Financial Stability Dashboard

Indicators	2015	2016	2017					
			Q4	Q1	Q2	Q3	Oct	Nov
1. Financial market sector								
Bond market								
Bond spread (10 years - 2 years)	1.1	0.6	0.8	1.1	1.1	1.0	1.0	1.0
Equity market								
SET index (end of period)	1,288.0	1,542.9	1,542.9	1,575.1	1,574.7	1,673.2	1,721.4	1,697.4
Actual volatility of SET index[1/]	13.9	15.2	15.2	7.0	4.8	5.8	9.3	8.5
Price to Earnings ratio (P/E ratio) (times)	22.6	18.6	18.6	17.4	16.3	17.9	18.6	18.3
Exchange rate market								
Actual volatility of Thai baht (%annualized)[2/]	5.1	4.4	5.0	3.5	3.9	2.9	3.0	2.6
Nominal Effective Exchange Rate (NEER)	108.5	106.2	107.0	108.7	109.8	111.2	112.0	113.1
Real Effective Exchange Rate (REER)	104.4	100.6	101.1	102.1	102.7	104.0	105.0	n.a.
2. Financial institution sector[3/]								
Minimum Lending Rate (MLR)[4/]	6.50	6.26	6.26	6.26	6.20	6.20	6.20	6.20
12-month fixed deposit rate[4/]	1.40	1.38	1.38	1.38	1.38	1.38	1.38	1.38
Capital adequacy								
Capital funds / Risk-weighted asset (%)	17.4	18.0	18.0	17.8	17.9	18.4	n.a.	n.a.
Earning and profitability								
Net profit (billion, Thai baht)	192.3	199.2	47.4	51.2	49.0	46.5	n.a.	n.a.
Return on assets (ROA) (times)	0.9	1.1	1.2	1.2	1.1	1.0	n.a.	n.a.
Liquidity								
Loan to Deposit and B/E (%)	97.0	96.3	96.3	95.7	96.5	96.4	n.a.	n.a.
3. Household sector								
Household debt to GDP (%)	81.2	79.9	79.9	78.7	78.4	n.a.	n.a.	n.a.
Financial assets to debt (times)	2.6	2.6	2.6	2.7	2.7	n.a.	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)								
Consumer loans	2.6	2.7	2.7	2.8	2.7	2.7	n.a.	n.a.
Housing loans	2.4	2.9	2.9	3.2	3.1	3.3	n.a.	n.a.
Auto leasing	2.3	1.8	1.8	1.6	1.7	1.6	n.a.	n.a.
Credit cards	4.0	3.7	3.7	3.8	3.2	2.6	n.a.	n.a.
Other personal loans	2.7	2.9	2.9	2.9	2.6	2.7	n.a.	n.a.
4. Non-financial corporate sector[5/]								
Operating profit margin (OPM) (%)	7.4	8.3	7.7	8.5	7.5	8.6	n.a.	n.a.
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.7	0.7	n.a.	n.a.
Interest coverage ratio (ICR) (times)	5.7	6.6	6.8	6.2	6.2	6.6	n.a.	n.a.
Current ratio (times)	1.7	1.6	1.6	1.7	1.7	1.7	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)								
Large businesses	1.6	1.5	1.4	1.6	1.8	1.7	n.a.	n.a.
SMEs	3.5	4.3	4.3	4.5	4.4	4.6	n.a.	n.a.

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2015	2016	2017					
			Q4	Q1	Q2	Q3	Oct	Nov
5. Real estate sector								
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)								
Total	59,667	61,452	16205	12244	15,086	16,992	3,915	n.a.
Single-detached and semi-detached houses	13,152	13,409	3179	2802	3,544	3,768	701	n.a.
Townhouses and commercial buildings	19,210	20,187	4967	4315	4,947	5,630	1,190	n.a.
Condominiums	27,305	27,856	8059	5127	6,595	7,594	2,024	n.a.
Number of new housing units launched for sale (Bangkok and Vicinity) (units)								
Total	107,988	110,575	31,452	25,304	25,529	34,369	2,211	n.a.
Single-detached and semi-detached houses	17,637	19,433	4,973	2,054	2,413	4,432	-	n.a.
Townhouses and commercial buildings	27,518	32,792	7,861	10,413	7,102	9,155	257	n.a.
Condominiums	62,833	58,350	18,618	12,837	16,014	20,782	1,954	n.a.
Housing price index (2009 = 100)								
Single-detached houses (including land)	129.3	128.8	128.8	128.6	129.6	131.6	131.2	n.a.
Townhouses (including land)	137.5	135.6	135.6	138.3	140.0	142.6	142.6	n.a.
Condominiums	160.9	173.6	173.6	169.8	168.8	169.8	170.8	n.a.
Land	168.8	171.3	171.3	171.3	164.2	172.9	173.3	n.a.
6. Fiscal sector								
Public debt to GDP (%)	43.9	41.2	41.2	42.2	41.8	42.4	41.8	n.a.
7. External sector								
Current account balance to GDP (%)[6/]	8.1	11.9	10.3	13.9	6.8	12.0	n.a.	n.a.
External debt to GDP (%)[7/]	32.0	32.7	32.7	33.5	34.3	36.0	n.a.	n.a.
External debt (billion, U.S. dollars)	131.1	132.2	132.2	136.2	140.5	148.9	145.5	n.a.
Short-term (%)	40.1	41.2	41.2	40.5	39.5	41.4	39.7	n.a.
Long-term (%)	59.9	58.8	58.8	59.5	60.5	58.6	60.3	n.a.
International reserves / Short-term external debt (times)	3.0	3.2	3.2	3.3	3.3	3.2	3.5	n.a.

Note:

[6/] Current account / Nominal GDP at the same quarter

[7/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2017		2018			2019		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
> 9	0	0	0	1	1	2	2	3
8-9	0	0	1	2	2	3	3	4
7-8	1	2	3	4	5	5	6	6
6-7	6	8	7	8	9	9	9	9
5-6	22	18	14	13	13	13	13	12
4-5	36	26	20	18	17	16	15	14
3-4	26	24	22	19	17	16	15	14
2-3	8	15	17	16	15	14	13	13
1-2	1	6	10	11	10	10	10	10
0-1	0	1	4	6	6	6	7	7
(-1)-0	0	0	1	3	3	3	4	4
(-2)-(-1)	0	0	0	1	1	1	2	2
(-3)-(-2)	0	0	0	0	0	1	1	1
< (-3)	0	0	0	0	0	0	0	1

Table: Probability distribution of headline inflation forecast

Percent	2017		2018			2019		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
> 4.0	0	0	2	5	3	6	6	7
3.5-4.0	0	0	3	4	3	5	4	5
3.0-3.5	0	0	6	6	5	6	6	6
2.5-3.0	1	2	9	8	7	9	8	8
2.0-2.5	3	5	13	11	9	10	10	10
1.5-2.0	12	10	15	12	11	11	11	11
1.0-1.5	25	16	15	12	12	12	11	11
0.5-1.0	30	20	13	11	12	11	11	10
0.0-0.5	19	19	10	10	11	9	9	9
(-0.5)-0.0	7	14	7	8	9	7	8	7
(-1.0)-(0.5)	2	8	4	5	7	5	6	6
(-1.5)-(1.0)	0	4	2	3	5	4	4	4
(-2.0)-(-1.5	0	1	1	2	3	2	2	3
< -2.0	0	0	0	2	4	2	3	3

Table: Probability distribution of core inflation forecast

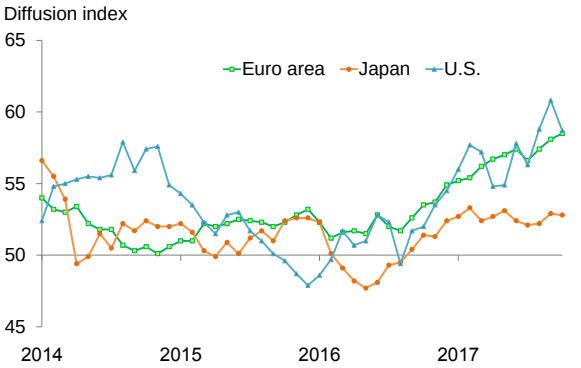
Percent	2017	2018				2019		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
> 4.0	0	0	0	0	0	0	0	0
3.5-4.0	0	0	0	0	0	0	0	0
3.0-3.5	0	0	0	0	0	0	1	1
2.5-3.0	0	0	0	1	1	2	2	3
2.0-2.5	0	0	2	4	4	6	7	8
1.5-2.0	0	2	9	12	12	14	15	16
1.0-1.5	8	19	26	23	22	23	23	22
0.5-1.0	57	44	34	28	26	25	23	22
0.0-0.5	33	29	22	20	20	18	17	16
(-0.5)-0.0	1	5	7	9	10	9	8	8
(-1.0)-(0.5)	0	0	1	3	3	3	3	3
(-1.5)-(1.0)	0	0	0	0	1	1	1	1
(-2.0)-(-1.5	0	0	0	0	0	0	0	0
< -2.0	0	0	0	0	0	0	0	0

4.2 Data pack

The Global Economy

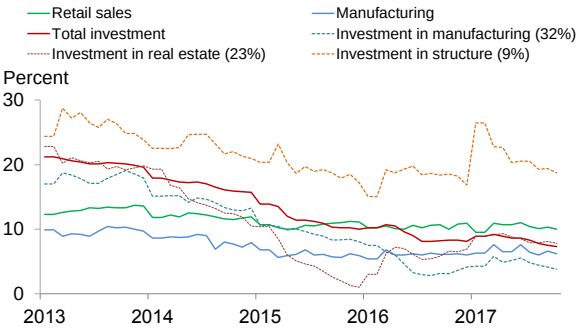
Thailand's trading partners economies continued to expand and remained a key driver of Thai exports going forward. However, a gradual increase of inflation allowed most central banks to maintain their accommodative monetary policy stance. Meanwhile, some central banks raised their policy rates such as the Bank of England and the Bank of Korea

Manufacturing Purchasing Manager Index



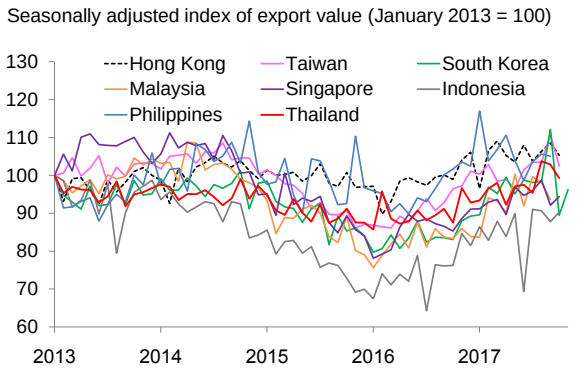
Source: Bloomberg

China's economic indicators (change from same period last year)



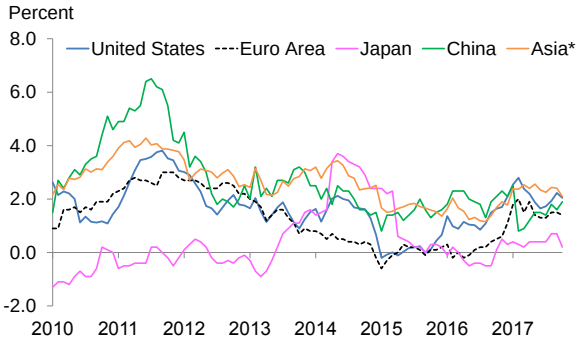
Note: ( ) denotes share to total investment
Source: CEIC

Asian exports



Source: CEIC

Inflation of Thailand's major trading partners

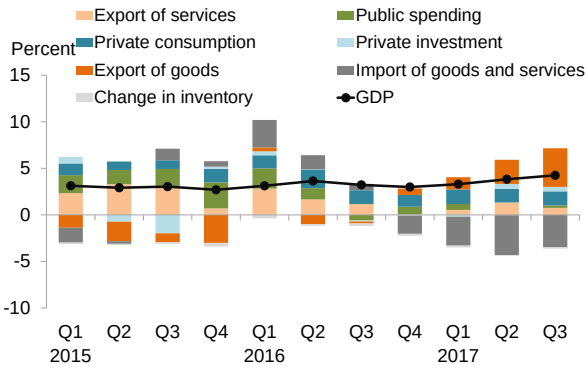


Note: * Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore and Taiwan
Source: CEIC

The Thai economy

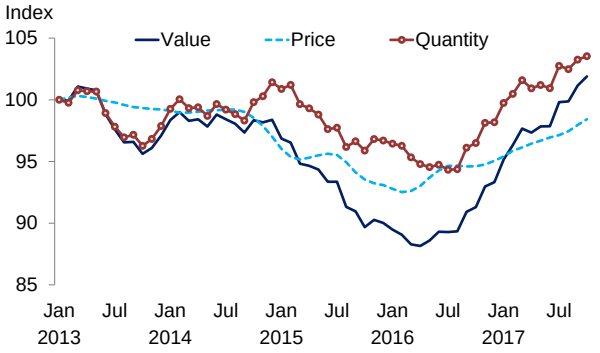
Thailand's economic growth gained further traction on the back of strong growth in merchandise exports and tourism as well as a continued domestic demand expansion. Public spending remained a key economic driver despite some contraction in public investment following prior acceleration in disbursement by state-owned enterprises.

Contribution to Thailand's GDP growth[1/]



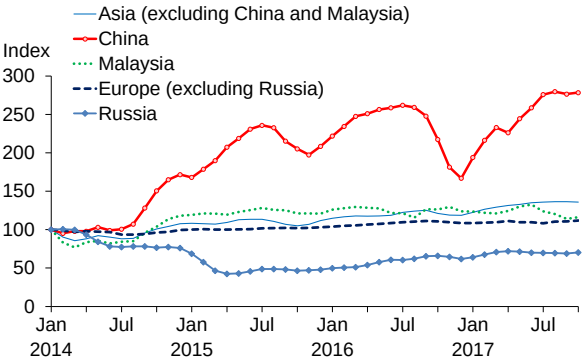
Note: [1/] Calculated by Chain Volume Measure method (CVM)
Source: Office of National Economic and Social Development Board and calculation by Bank of Thailand

Thai exports (excluding gold): value, price, and quantity
(seasonally adjusted 3-month moving average, January 2013 = 100)



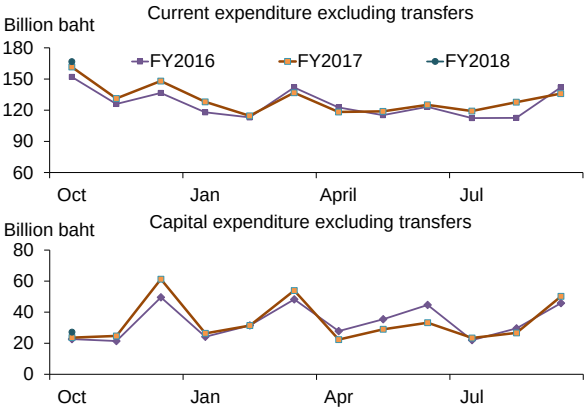
Source: Customs Department and Ministry of Commerce, and calculation by Bank of Thailand

Foreign tourists classified by nationality
(seasonally adjusted 3-month moving average, January 2014 = 100)



Source: Department of Tourism

Public spending by central government

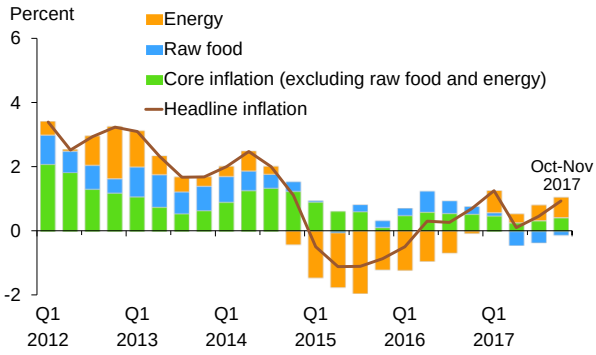


Source: Bureau of Budget, Fiscal Policy Office

Inflation

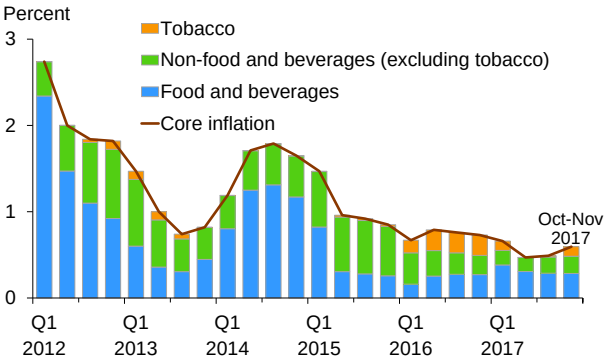
Headline inflation edged up mainly due to energy prices, while core inflation slightly increased on account of an excise tax increase. Meanwhile, five-year-ahead inflation expectations of professional forecasters declined.

Contribution to headline inflation



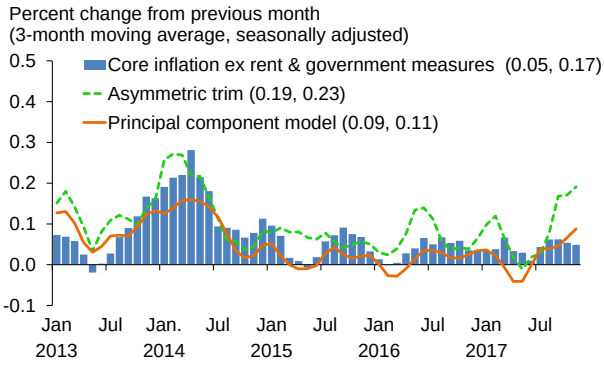
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, and calculation by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, and calculation by Bank of Thailand

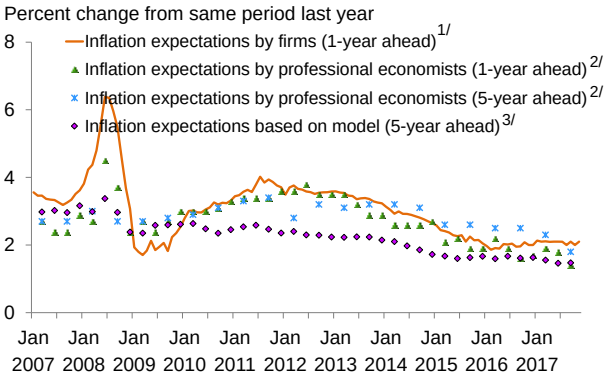
Underlying inflation indicators



Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of November 2017, while the second value is 2004 - 2014 average; **Asymmetric trim** excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; **Principal component model** calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, and calculation by Bank of Thailand

Inflation expectations

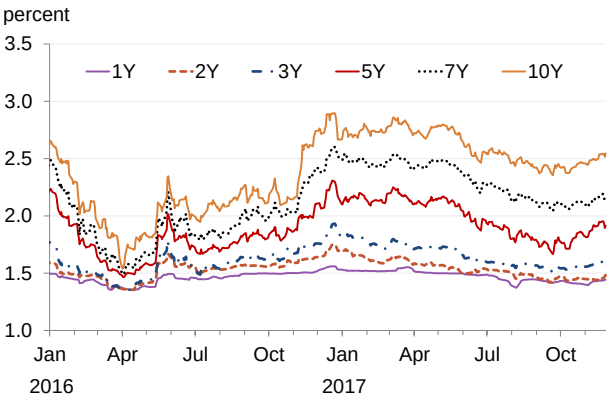


Sources: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

Financial conditions

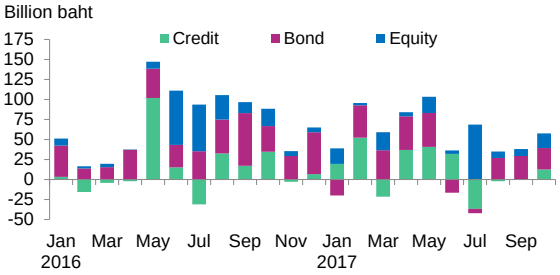
Short-term money market rates stayed low while long-term government bond yields rose due to both supply-side and external factors. Total corporate financing continued to expand. The Thai baht appreciated against the U.S. dollar due to external uncertainties in line with regional currencies. Meanwhile, the Nominal Effective Exchange Rate (NEER) strengthened somewhat consistent with improvements in economic fundamentals.

Thai government bond yields



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

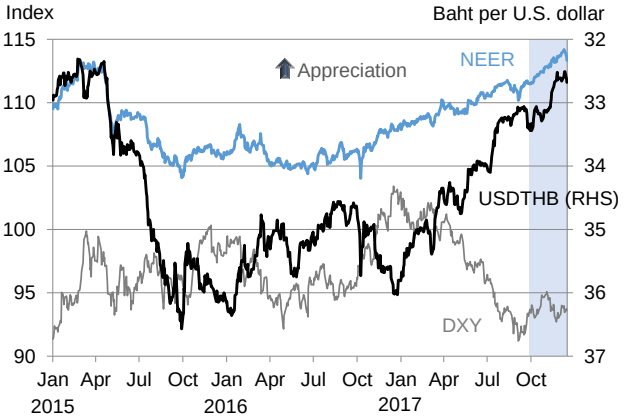
Total corporate financing by instrument*



Note: * Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bonds excluding commercial banks, and newly issued equities.

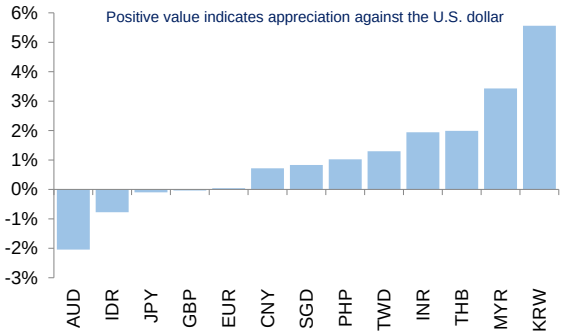
Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), Dollar Index (DXY)



Sources: Bank of Thailand and Reuters (data as of 19 December 2017)

Currency movements vis-a-vis U.S. dollar (19 Dec 2017 compared to 30 Sep 17)

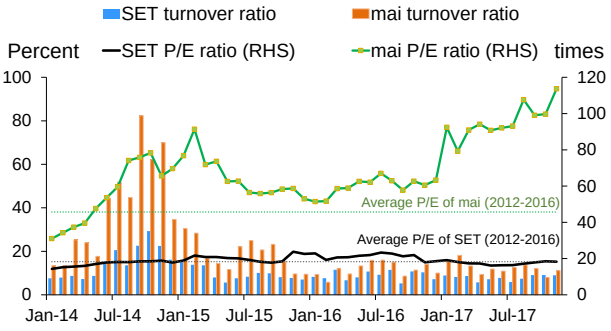


Sources: Bank of Thailand and Reuters (data as of 19 Dec 2017)

Stability: financial markets

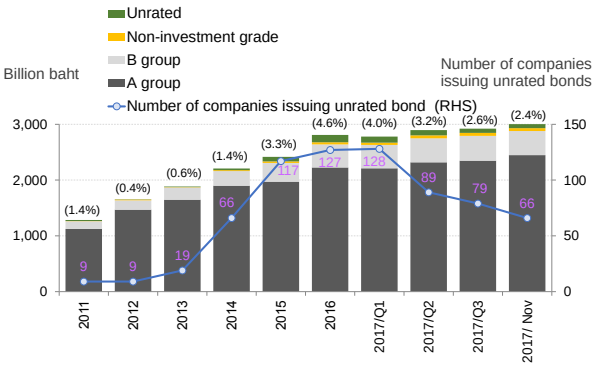
The price-to-earning (P/E) ratio of the Stock Exchange of Thailand stayed close to the historical average. The P/E ratio of the Market for Alternative Investment (mai) remained high due to declining business performance in the third quarter of 2017, especially in agriculture, trade and services sectors. New issuance of unrated bonds continued to decline since the end of 2016 after some defaults. As a result, the ratio of unrated bonds to total corporate bonds outstanding continued to fall.

Current price-to-earning ratio and turnover ratio of SET and mai



Source: Stock Exchange of Thailand (as of November 2017)

Corporate bonds outstanding

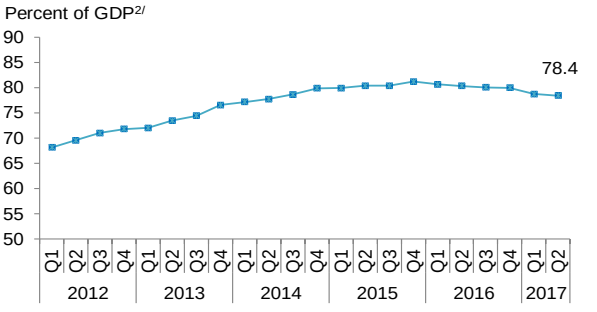


Note: ( ) represents percent of unrated bonds in total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Stability: household sector

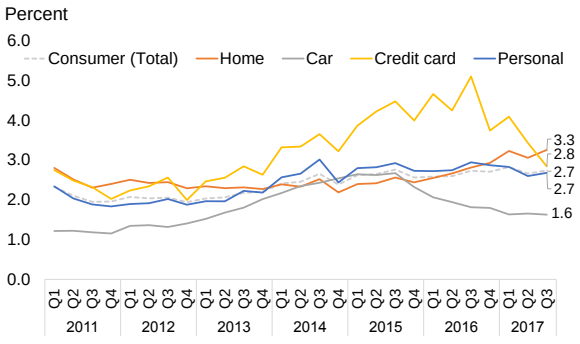
Household debt remained high, although the ratio of household debt to GDP continued to decline. Deleveraging remained concentrated among the high-income group, reflecting a recovery of the economy that was not yet broad-based. However, deterioration in debt serviceability of households warranted close monitoring going forward.

Household debt[1/]



Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

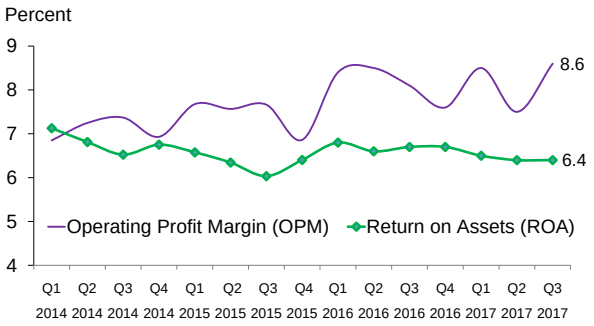


Source: Bank of Thailand

Stability: corporate sector

Overall stability of the corporate sector remained sound. Profitability and debt serviceability improved in line with economic conditions. However, the share of non-performing loans in total loans (NPL) of small businesses continued to increase.

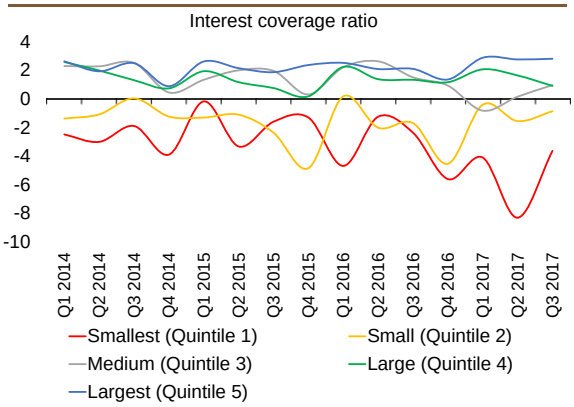
Operating profit margin (OPM) and return on assets (ROA)*



Note: * Median estimates; ROA is returns to average assets. OPM is operating profits to total sales.

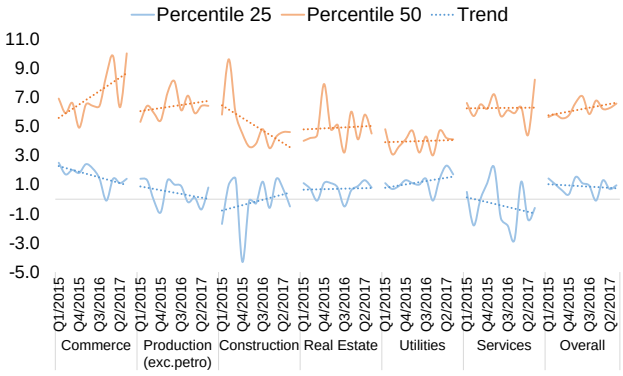
Source: Stock Exchange of Thailand and calculation by Bank of Thailand

Debt serviceability at 25th percentile classified by firm size



Source: Stock Exchange of Thailand and calculation by Bank of Thailand

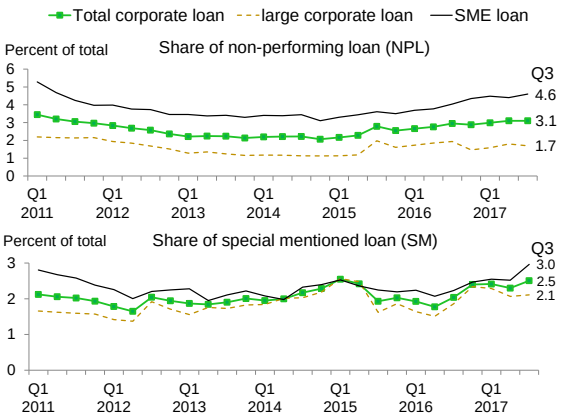
Interest coverage ratio classified by sector



Note: * production exclude Petroleum and chemicals

Source: Stock Exchange of Thailand and calculation by Bank of Thailand

Loan quality of corporate sector

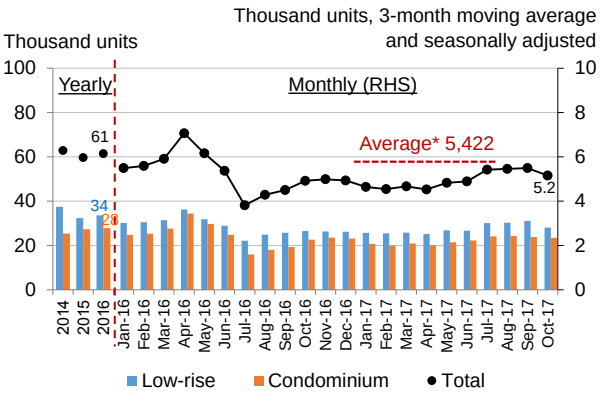


Source: Bank of Thailand

Stability: real estate sector

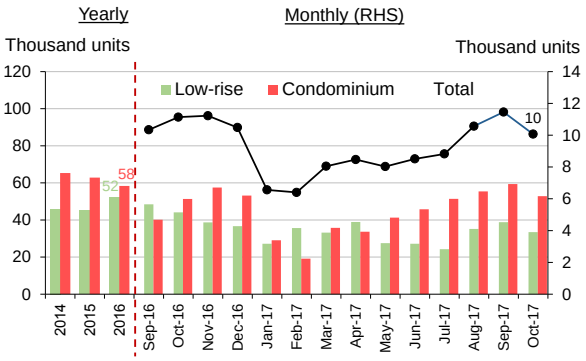
The real estate sector overall slowed down from the previous quarter, partly due to lowered demand and supply from temporary factors in October 2016. Meanwhile, residential property prices remained broadly stable.

Residential units in Bangkok and vicinity



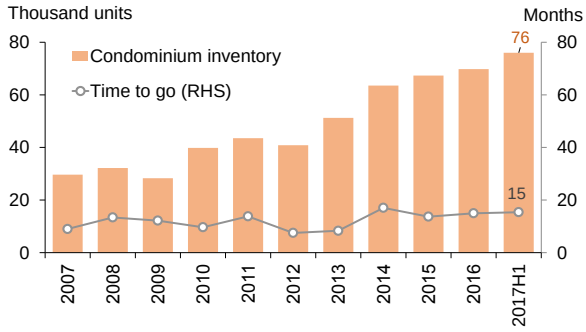
Note: *Average during 2010-2016, excluding periods with government's stimulus measures (November 2015 - April 2016)
Source: Bank of Thailand

New residential projects launched in Bangkok and vicinity



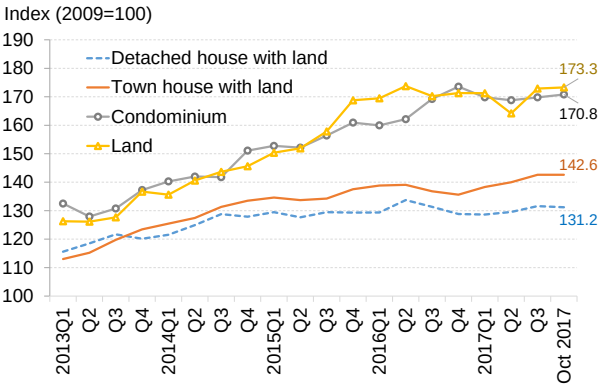
Source: Agency for Real Estate Affairs (AREA) and calculation by Bank of Thailand

Condominium inventory in Bangkok and vicinity and 'time to go'



Note: 'Time to go' is the time taken for all real estate inventory to be sold, using the average sales rate over the past 12 months.
Source: AREA and calculation by Bank of Thailand

Real estate prices

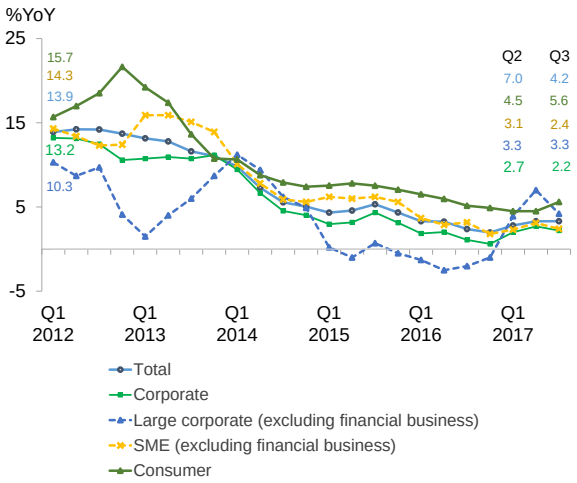


Source: Bank of Thailand

Stability: financial institutions

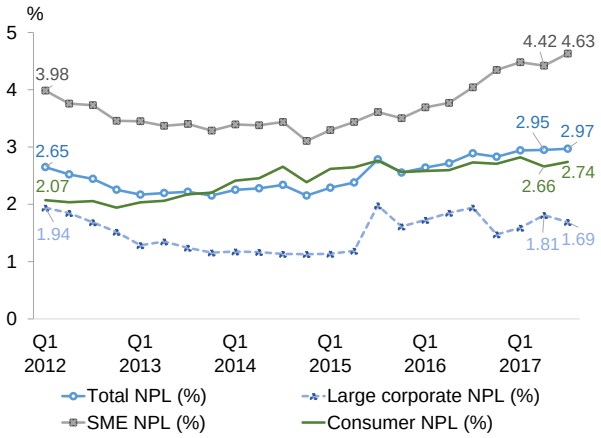
Credit growth and the NPL ratio in the third quarter remained close to the previous quarter, while credit quality of SMEs and mortgage loans warranted monitoring. Nonetheless, the financial system remained sound with high levels of provisions and capital buffers.

Commercial bank credit growth



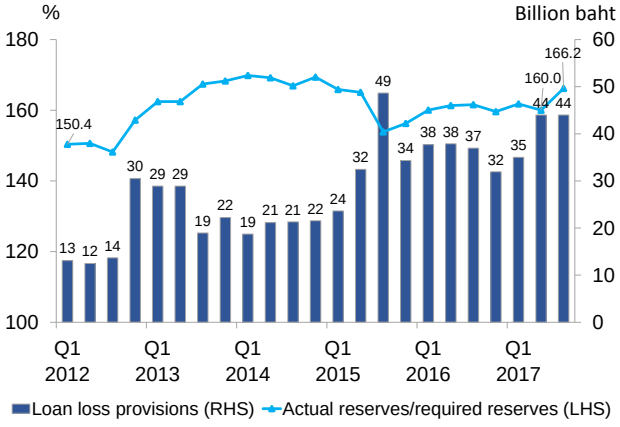
Source: Bank of Thailand

Non-performing loans (NPL)



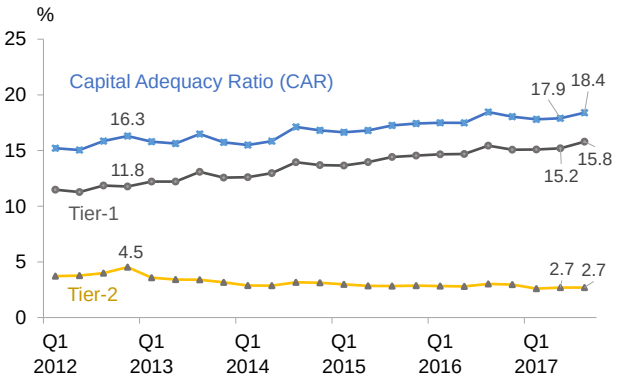
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

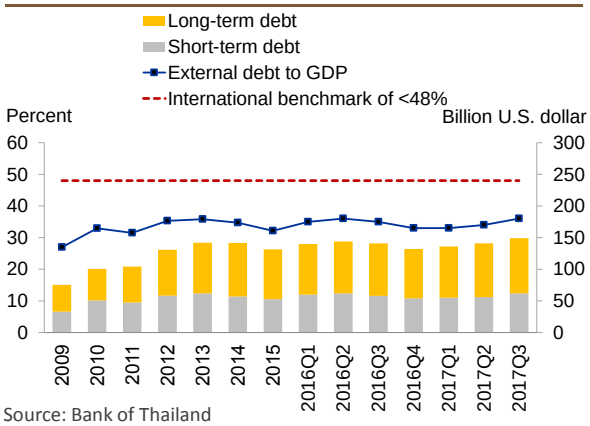


Source: Bank of Thailand

Stability: External position

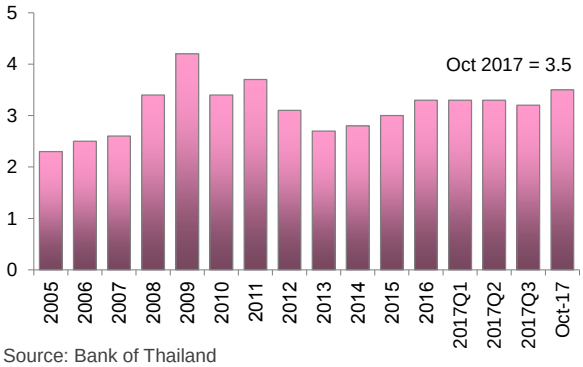
Thailand's external stability remained strong due to a lower level of external debt than an international benchmark, with international reserves at a high level relative to short-term debt.

Thailand's external debt



Source: Bank of Thailand

Ratio of international reserves to short-term debt

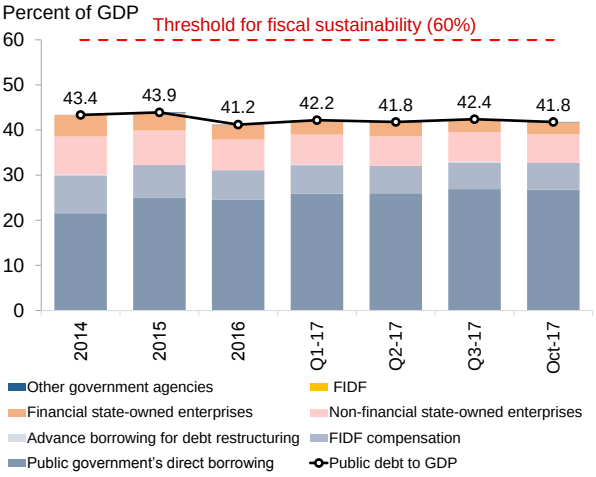


Source: Bank of Thailand

Stability: fiscal sector

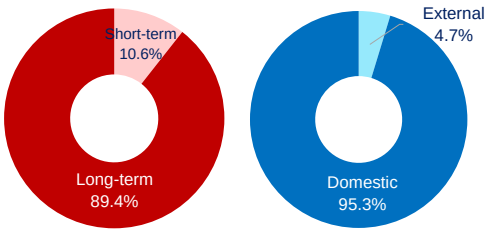
Fiscal stability remained sound. The ratio of public debt to GDP stayed below the sustainability threshold.

Ratio of public debt to GDP



Note: Calculated by GDP with Chain Volume Measure
Source: Public Debt Management Office

Outstanding debt as of October 2017



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office

Pursuing Sustainable
Economic Well-Being

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